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Cross-species Standardised Cortico-Subcortical Tractography

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eLife Assessment

This **important** study provides a novel approach for delineating subcortical-cortical white matter bundles. The authors provide **convincing** evidence by harnessing state-of-the-art methods and cross-species data. Together, this effort will be of interest to scientists across multiple subfields and accelerate progress in a biologically critical but methodologically challenging area.

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Abstract

Despite their importance for brain function, cortico-subcortical white matter tracts are underrepresented in diffusion MRI tractography studies. Their non-invasive mapping is more challenging and less explored compared to other major cortico-cortical bundles. We introduce a set of standardised tractography protocols for delineating tracts between the cortex and various deep subcortical structures, including the caudate, putamen, amygdala, thalamus and hippocampus. To enable comparative studies, our protocols are designed for both human and macaque brains. We demonstrate how tractography reconstructions follow topographical principles obtained from tracers in the macaque and how these translate to humans. We show that the proposed protocols are robust against data quality and preserve aspects of individual variability stemming from family structure in humans. Lastly, we demonstrate the value of these species-matched protocols in mapping homologous grey matter regions in humans and macaques, both in cortex and subcortex.

1 Introduction

Function-specific brain activity involves the integration of information from multiple remote brain regions. This integration is enabled by white matter (WM) bundles interconnecting different brain regions (1, 2, 3, 4). Of particular interest and importance are bundles connecting cortical areas with deep brain structures. Subcortical structures have important roles in affective, cognitive, motor and social functions (5,6,7), which emerge through interactions with cortical areas that such connections enable (8, 9, 10). Hence, the variability of these connections between individuals has been linked to differences in behavioural traits (11, 12, 13). Furthermore, their disruption has been associated with abnormal function and pathology in neurodegenerative and mental health disorders (14, 15, 16, 17). In the clinic, individual variability in cortico-subcortical connectivity has been used to assist presurgical planning and predict personalised targets for efficacious interventions (18).

Chemical tracer studies in the non-human primate (NHP) brain, have provided, and continue to provide, invaluable insights into cortico-subcortical connections, and neuroanatomy in general (19). Examples include tracing of cortico-striatal connections (20, 21, 8, 22), amygdalofugal connections (23) and thalamo-cortical connections (24,20). Comparative neuroanatomy studies can subsequently explore and translate principles of white matter organisation learnt from NHPs to humans (25, 22, 26)). Brain imaging and, in particular, diffusion magnetic resonance imaging (dMRI) tractography (27) is a crucial component in these comparative studies and beyond (28, 29, 30), allowing non-invasive mapping of these connections in the living human.

Towards this direction, recent dMRI-based frameworks have been developed to map respective white matter bundles ("tracts") across NHPs and humans (31, 32, 33, 34, 29). These rely on standardised dMRI tractography protocols, comprising of functionally-driven, rather than geometric, definitions, enabling automated and generalisable mapping of homologous major white matter tracts across species (31). These developments have allowed cross-species neuroanatomy studies (35, 36) and mapping of connections across humans to study links with brain development, function and dysfunction (37, 38). A current limitation of these imaging-based approaches is that they have mainly focused so far on cortico-cortical bundles (with the exception of cortico-thalamic connections).

Tractography protocols for white matter bundles that reach deeper subcortical regions, for instance the striatum or the amygdala, are more difficult to standardise. The relative size and proximity of these bundles, and the white matter complexities and bottlenecks they go through, can make their mapping through dMRI particularly challenging. As a consequence, considerably fewer studies have proposed solutions for their reproducible reconstruction, both within and across primate species, compared to more major cortico-cortical bundles (39, 40, 31, 41). Some existing studies have focused on cortico-striatal bundles (42,43), uncinate and amygdalofugal fasciculi (26), parts of the extreme capsule (44) and the anterior limb of the internal capsule (25,22). However, these either utilise labour-intensive single-subject protocols (22,26), are not designed to be generalisable across species (42, 43), or are based mostly on geometrically-driven parcellations that do not necessarily preserve topographical principles of connections (40). We propose an approach that addresses these challenges and is automated, standardised, generalisable across two species and includes a larger set of cortico-subcortical bundles than considered before, yielding tractography reconstructions that are driven by neuroanatomical constraints.

Specifically, we build upon our previous work on FSL-XTRACT (35, 31, 29) to propose standardised protocols and an end-to-end framework for automated cortico-subcortical tractography in the macaque and human brain, considering connections between the cortex and the caudate, putamen and amygdala. To this end, we use prior anatomical knowledge from NHP tracers to define new generalisable protocols, including the amygdalofugal tract, the Muratoff bundle and the striatal bundle (external capsule) with its frontal, sensorimotor, temporal and parietal parts, augmenting our previous protocols for hippocampal and thalamic tracts (31). Due to their close proximity, we also develop new protocols for the respective extreme capsule parts (frontal, temporal, parietal) and revise previously released protocols (31) for the uncinate fasciculus, anterior commissure and fornix.

We demonstrate the mapping of the respective bundles in the human and macaque brain and show that tractography reconstructions follow topographical principles obtained from tracers. We show that the proposed definitions are robust against dMRI data quality and preserve aspects of individual variability stemming from family structure in humans, as reflected by higher similarity of reconstructed tracts in the brains of monozygotic twins compared to non-twin siblings and unrelated subjects. We subsequently demonstrate how these tractography reconstructions can improve the identification of homologous grey matter (GM) regions across species, both in cortex and subcortex, on the basis of similarity of grey matter areal connection patterns to the set of proposed white matter bundles (1, 35).

2 Results

Using prior anatomical knowledge from tracer studies in the macaque, we developed new tractography protocols for the macaque brain and subsequently translated them to the human brain. We considered 23 tracts in total (11 bilateral, 1 commissural), which included tracts connecting the cortex to the amygdala, caudate and putamen. Specifically, we developed protocols for the amygdalofugal (*AMF*) pathway, the sensorimotor, frontal, temporal and parietal parts of the striatal bundle/external capsule (StB_m, StB_f, StB_t, StB_p), and the Muratoff bundle (*MB*). Due to their proximity, we also developed protocols for the frontal, temporal and parietal parts of the extreme capsule (EmC_f, EmC_t, EmC_p) (neighbouring to the corresponding external capsule parts), and revised previous protocols for the uncinate fasciculus (*UF*) (neighbouring to the *AMF*), the fornix (*FX*) (output tract of the hippocampus next to the amygdala), as well as the anterior commissure (*AC*) (**Table 1** and Supplementary Table S1).

We used the XTRACT approach (31) to define tractography protocols, governed by two principles: (i) protocols are comprised of seed/stop/target/exclusion regions of interest (ROIs) defined in template space, so that they are standardised and generalisable (compared to subject-specific protocols), (ii) ROIs are coarse enough and defined equivalently between macaques and humans to enable the tracking of corresponding bundles across species. Full tractography protocols, and modifications to existing protocols, are described in detail in Methods. Protocols were defined in MNI152 template space for human tractography and F99 space (45, 46) for macaque tractography. Additionally, we generalised the macaque protocols to the NIMH Macaque Template (NMT v2) (47). For clarity, results shown in the main text use the F99 space protocols. Comparison between results in NMT and F99 space can be seen in Supplementary Figure S3.

2.1 Subcortical tract reconstruction across species and comparisons with tracers

Using dMRI data from the macaque ($N = 6$) and human brain ($N = 50$) and the defined protocols, we performed tractography reconstructions for all the tracts of interest. Maximum intensity projections of the resultant group-averaged tract reconstructions for the macaque and human are

Tract Name	Abbreviation
Amygdalofugal Tract	AMF
Anterior Commissure	AC
Extreme Capsule (frontal)	EmC_f
Extreme Capsule (temporal)	EmC_t
Extreme Capsule (parietal)	EmC_p
Fornix	FX
Muratoff Bundle	MB
Striatal Bundle (sensorimotor)	StB_m
Striatal Bundle (frontal)	StB_f
Striatal Bundle (temporal)	StB_t
Striatal Bundle (parietal)	StB_p
Uncinate Fasciculus	UF

Table 1

New and Revised Subcortical Protocols.

The developed subcortical tractography protocols for the macaque and human brain. Protocols for Anterior Commissure, Fornix and Uncinate Fasciculus were revised from (31).

shown colour-coded in **Figure 1** (individual tracts can be seen in Supplementary Figure S1). These reveal overall correspondence in the main bodies of tracts across species, whilst capturing differences in (sub)cortical projections.

We explored whether white matter organisation principles known from the tracer literature are captured in these tractography reconstructions (48,49,20,50,51,10). For instance, the striatal bundle (*StB*)/external capsule is always medial to the extreme capsule and the Muratoff bundle runs along the head of the caudate nucleus. **Figure 2A** shows the relative positioning for StB_f, EmC_f and *MB* bundles. Correspondence between tractography results and tract tracing reconstruction in the macaque can be observed, with their relative positions being preserved. This relative position was preserved in the human tractography results as well. Furthermore, the medio-lateral separation is also observed in the other parts of *StB* and *EmC* (i.e. parietal, temporal), as shown in Supplementary Figure S2. Similarly for the amygdalofugal (*AMF*) bundle (**Figure 2B**), this runs through the anterior commissure and ventral pallidum, as well as, in its lateral part, over the uncinate fasciculus (*UF*). We see agreement with respect to these relative positions in both the macaque and human.

In addition to the main white matter core of the reconstructed bundles, we also explored agreement of the relative connectivity patterns within the striatum between tracers and tractography. Cortical injections of anterograde tracers from different parts of the macaque brain reveal a dorsolateral to ventromedial organisation in the putamen, from parietal to temporal projections (**Figure 3**). Using the path distribution of the tractography reconstructed *StB* parts within the putamen, we could obtain a similar pattern in the macaque brain. This also resembled the pattern found in the human brain, as shown in both coronal and axial views.

2.2 Generalisability across data and individuals

We subsequently explored generalisability and robustness of the tractography protocols against NHP template spaces and dMRI data quality. Supplementary Figure S3 shows tract reconstructions in the macaque brain when using the F99 (45) vs the NMT (47) templates, similar tractography reconstructions for protocols defined in either of the two templates.

To explore performance against data quality, we compared tractography reconstruction in very high-quality high-resolution data from the Human Connectome Project (HCP) (57, 28), to tractography in more standard quality data from the UK Biobank dataset (58). Supplementary Figure S4 demonstrates the ability to reconstruct all tracts across a range of data qualities, with good correspondence of the main bodies of the tracts in both datasets. We quantified this agreement by calculating the mean Pearson's correlation across the set of new and revised tracts for each unique pair of subjects across and within each of the HCP and UK Biobank (UKB) datasets (**Figure 4A**). For reference, we performed similar correlations for the original set of XTRACT tracts (31) (see Supplementary Table S1 for a list of Original vs New+Revised tracts). Higher correlation was observed within each dataset, but also a sufficiently high correlation between the two datasets. We found similar patterns across datasets both for the original and the new tracts, showcasing that the new protocols behave similarly to the widely-used original XTRACT protocols, across data qualities.

The mean agreement between HCP and UKB reconstructions was lower compared to withindataset agreements. The two cohorts correspond to different age ranges, with HCP having younger adults than the UKB, which could be contributing to these differences. In addition, this was due to occasionally reconstructing a sparser path distribution in the low-resolution data, particularly for some of the new tracts, as both their relative size and their proximity make them more challenging. This highlights the potential importance of having high-resolution data in tracking white matter bundles in densely-packed areas of higher complexity. It is interesting to note, however, that similar tracts were less/more reproducible between subjects across data

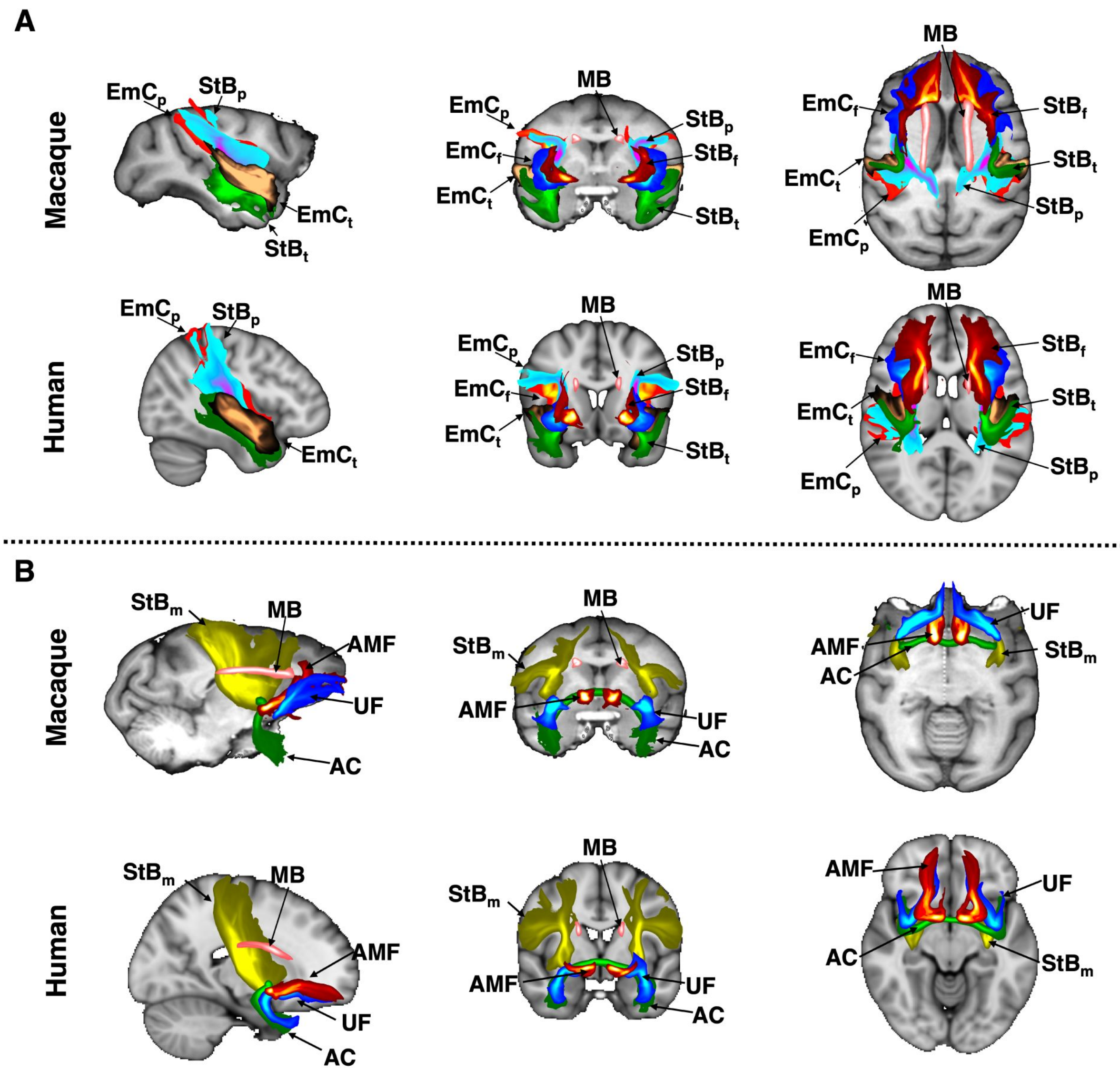

Figure 1

Tractography reconstructions of subcortical bundles in the macaque and human brain using correspondingly defined protocols.

Maximum intensity projections (MIPs) in sagittal, coronal and axial views of group-averaged probabilistic path distributions, for all proposed tractography protocols in the macaque (6 animal average) and human (average of 50 subjects from the Human Connectome Project). All MIPs are within a window of 20% of the field of view centred at the displayed slices. (A) Frontal, temporal and parietal parts of the extreme capsule (EmC_f, EmC_t, EmC_p); frontal, temporal and parietal parts of the striatal bundle (StB_f, StB_t, StB_p); and the Muratoff bundle (MB). (B) amygdalofugal tract (AMF); anterior commissure (AC); uncinate fasciculus (UF); sensorimotor part of the striatal bundle (StB_m); Muratoff bundle (MB). Path distributions were thresholded at 0.1% before averaging.

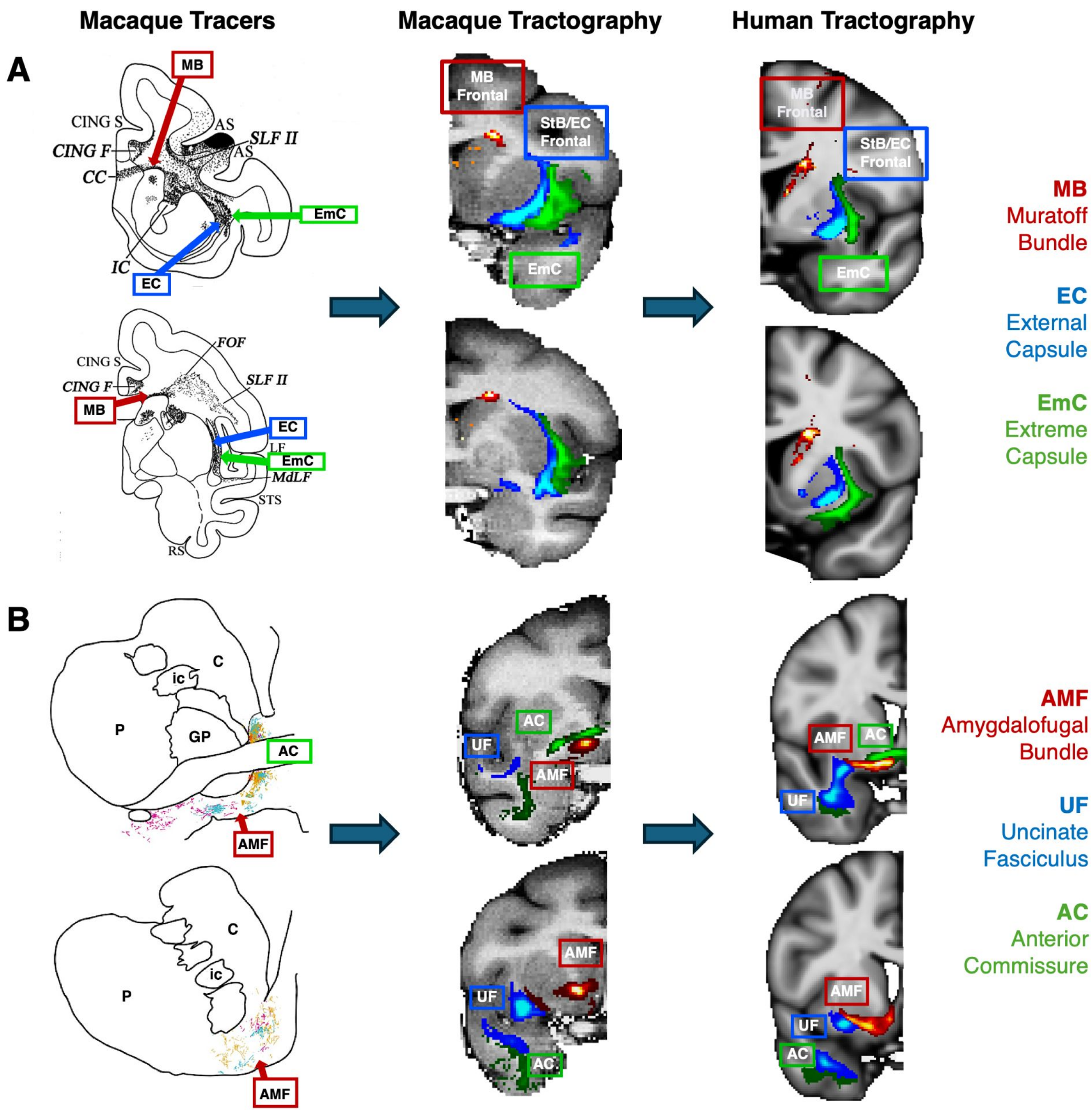

Figure 2

Tractography mirrors tracer patterns in the macaque brain, with similar patterns in the human.

The proposed protocols were first developed in the macaque guided by tracer literature, and then transferred over to the human. Relative positioning of dMRI-reconstructed tracts was subsequently explored against the ones suggested by tracers, with good agreement in both species. **(A)** The dorsal-medial/ventral-lateral separation between the extreme an external capsule (here the frontal parts EmC_f and StB_f shown) is present in macaque tractography, as suggested in the tracer literature. The Muratoff bundle runs along the head of the caudate nucleus. These relative positions are also preserved in the human tractography results. Tracer image modified from (52) with permission. **(B)** Similarly for the amygdalofugal bundle (AMF), which runs under the Anterior Commissure (AC) and over the Uncinate Fasciculus (UF), we see agreement with tracer studies with respect to its location in both the macaque and human tractography (23, 26, 53). Tracer image adapted from (23) with permission. In all examples group-average tractography results are shown.

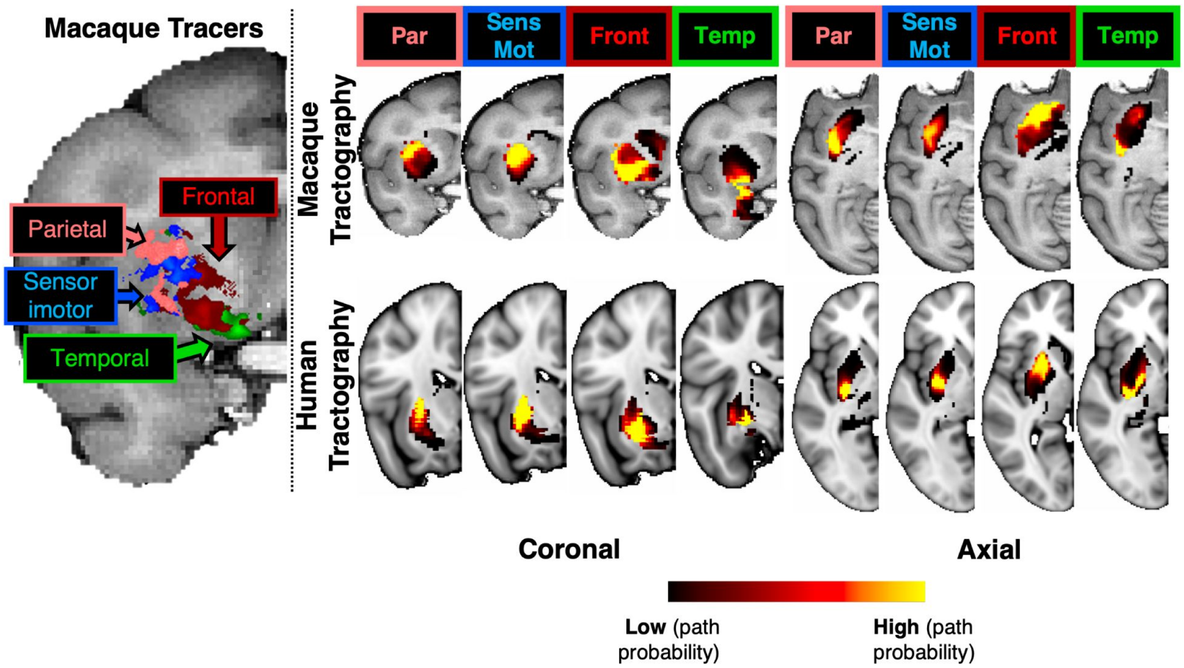

Figure 3

Tractography-derived connectivity patterns in the putamen resemble (for both macaque and human) termination sites identified by tracers after injections at different cortical areas (frontal, sensorimotor, parietal, temporal) in the macaque.

Left: Using macaque tracer data from 78 injections in various parts of the cortex, tracer termination sites in the putamen suggested a pattern based on the distinct cortical origin of the tracer injection sites; moving from the dorsolateral to the ventromedial putamen. Right: The path distributions of the different parts of the striatal bundle (StB_f, StB_m, StB_p, StB_t) within the putamen reveal a similar pattern of connectivity to different parts of the cortex, both for macaque (top) and the human (bottom). Coronal and axial views of group-average results are shown for tractography. Cortical areas (Front: frontal cortex, Par: parietal cortex, Temp: temporal cortex, SensMot: Sensorimotor cortex) were obtained from the CHARM1 parcellation (54) in the macaque brain (for both tracers and tractography) and from the Harvard parcellation in the human (55, 56, 49).

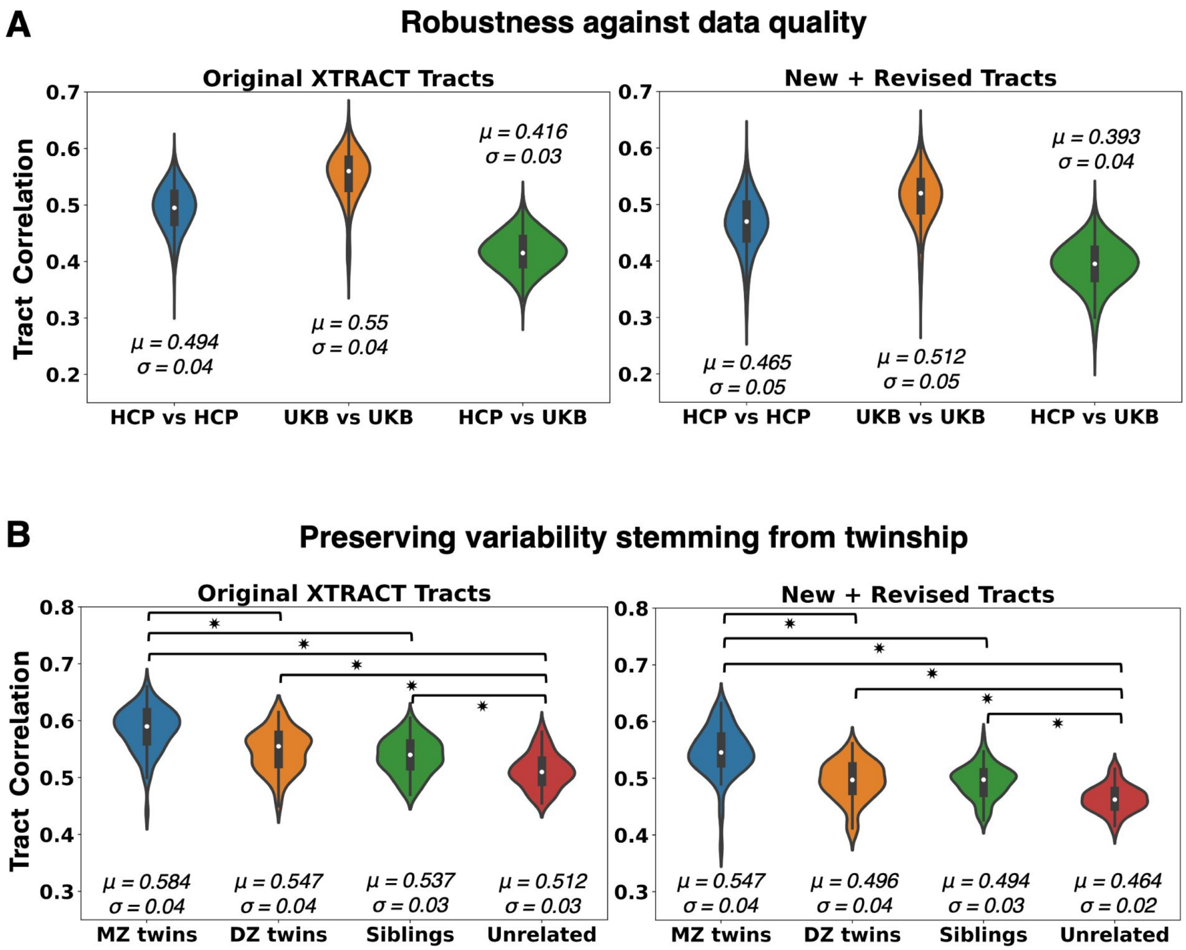

Figure 4

Generalisability of proposed tractography reconstructions across data quality and individuals.

Results for the new subcortical tracts (right column) are shown against reference corresponding results for the original set of XTRACT tracts (left column), which have been widely used (31). (**A**) Tract similarity within and between two in-vivo human cohorts, spanning a wide range of dMRI data quality (HCP: high resolution, long scan time, bespoke setup, UK Biobank (UKB): standard resolution, short scan time, clinical scanner). Violin plots of the average across tracts pairwise Pearson's correlations, between 1,225 unique subject pairs within and across the two cohorts, are shown. Correlations are performed on normalised tract density maps with a threshold of 0.5%. Reported μ is the mean of the correlations across tracts and subject pairs and σ is the standard deviation. (**B**) Tract similarity in twins, non-twin siblings and unrelated subjects. Violin plots of the average across tracts pairwise Pearson's correlations between 72 monozygotic (MZ) twin pairs, 72 dizygotic (DZ) twin pairs, 72 non-twin sibling pairs, and 72 unrelated subject pairs from the Human Connectome Project. Heritable traits are more similar in MZ twins, equally similar in DZ twins and non-twin siblings and more than in unrelated subjects. Asterisk indicates significant pairwise comparisons between groups, as indicated by the brackets.

qualities. For the lower quality data from the UKB, the tracts with lowest agreement across subjects (4) were the anterior commissure (AC) and the temporal part of the Extreme Capsule (EmC_t), while the highest correlations were for the Muratoff Bundle (MB) and the temporal part of the Striatal Bundle (StB_t). For the higher quality HCP data, the temporal part of the Extreme Capsule (EmC_t) and the Muratoff Bundle (MB) were also the tracts with the lowest/highest correlations across subjects, respectively. Hence, certain tract reconstructions were consistently more variable than others across subjects, which may hint to also being more challenging to reconstruct. Taken together, despite differences, our results suggest all tracts could be reconstructed across both data qualities in a generalisable manner (Supplementary Figure S4).

We subsequently explored whether the proposed protocols preserve aspects of individual variability. We used the family structure in the HCP data to explore whether tract reconstructions from monozygotic twin pairs are more similar compared to tracts obtained from other pairs of siblings or unrelated subjects. As shown in **Figure 4B**, we found a decrease in pairwise tract similarity going from monozygotic twins, to dizygotic twins and non-twin siblings, and to pairs of unrelated subjects. For reference, we performed the same analysis for the original XTRACT tracts and the same pattern persisted for the new (and revised) tracts, in agreement with previous work (59, 60, 31). In each analysis, all pairwise differences were significant (Bonferroni corrected $p < 0.05$; following a Mann-Whitney U-Test), with the exception of dizygotic twins compared to non-twin siblings.

Examples of tract reconstructions on individual subjects are shown in Supplementary Figures S6, S7. The figures demonstrate tractography results for subjects corresponding to 10^{th}, 50^{th} and 90^{th} percentile of the distribution of tract correlations to the HCP group average. This ranking was also representative of high, medium and low subject motion across the cohort respectively. Results demonstrate that the expected patterns are preserved for all tracts (MB, AMF, UF, EmC, StB (frontal and parietal parts)). Supplementary Figure S6 shows that even the relative medial-lateral organisation of the StB with respect to the EmC is also maintained across the three individual examples, in agreement with the group average pattern.

2.3 Identifying homologues in cortex and subcortex using tractography patterns

Based on our previous work (35), we used the similarity of areal connectivity patterns with respect to equivalently defined white matter tracts across the two species, to identify homologous grey matter regions between humans and macaques. With the addition of the new subcortical tracts we could perform this task for deep brain structures (subcortical nuclei and hippocampus) with considerably greater granularity than before. **Figure 5** demonstrates such identification task for five structures in the left hemisphere (caudate, putamen, thalamus, amygdala, hippocampus) using cortico-cortical and cortico-subcortical tracts (sets of tracts defined in Supplementary Table S1). On the left, the regions in the macaque brain with the lowest divergence (highest similarity) in their connectivity patterns to the connectivity patterns of the corresponding human regions are shown in blue. Using only connectivity pattern similarity, these five structures can be matched almost perfectly across the two species. For instance, human putamen (left hemisphere) has more similar connectivity (lower divergence) to macaque putamen (left hemisphere), human thalamus (left hemisphere) to macaque thalamus (left hemisphere), etc. Since we are mapping structures in the left hemisphere using left hemisphere tracts, we observe a low similarity in the contralateral (right) hemisphere, as expected. On the right of **Figure 5**, this identification is quantified even further, highlighting the value of considering the new tracts. For every human left-hemisphere region (specified on the vertical axis) the boxplot of divergence of connectivity patterns to each of the five macaque deep brain regions (left-hemisphere) is plotted. The best match corresponds to the boxplot with the lowest values (green) and the dashed blue lines show the medians of these boxplots for each case. For reference, the medians of the divergence values when not considering the new subcortical tracts are shown with the red dashed lines, which are overall more flat (with the exception of the hippocampus which has a connectivity pattern strongly driven by the dorsal subsection of the cingulum bundle (*CBD*), a cortico-cortical tract). It is evident that considering the new tracts provides enhanced contrast between the subcortical structures connectivity patterns, enabling their correct identification. The improvement is thus not in the best match, but in the specificity of the match.

Having shown increased contrast and specificity in the mapping of deep brain structures, we investigated whether we see a similar effect in the cortex. We selected a set of nearby frontal region pairs to map across the human and the macaque (**Figure 6**), since a number of the new tracts connect frontal regions to the subcortex. Specifically, we considered the dorsomedial prefrontal cortex (*dmPFC*), the ventromedial prefrontal cortex (*vmPFC*), the rostral orbitofrontal cortex (OFC_r), and the frontal operculum (*FOp*). These regions were also chosen as they are part of different functional networks (default mode, limbic, and frontoparietal networks), equivalently defined between the macaque and human (61).

The prediction from human to macaque is shown in **Figure 6A**, while the converse prediction from macaque to human is shown in **Figure 6B**. In each case, we compare the prediction using only cortico-cortical tracts (column 1), using only cortico-subcortical tracts (column 2), and using the full set (column 3) (sets of tracts defined in Supplementary Table S1 - the middle cerebellar peduncle (*MCP*) was not used in these comparisons). The predicted areas with the highest similarity in connectivity patterns are depicted in blue, while the a priori expected homologue region borders have been outlined in cyan. These results demonstrate benefits when using the subcortical tracts, with mapping of some regions (for instance *vmPFC* and OFC_r) being improved more than others. However, in general, we observed an increase in cross-species similarity in the corresponding areas of interest, combined with a decrease in similarity everywhere else in the cortex, when we considered cortico-subcortical tracts (columns 2, 3) compared to when we considered cortico-cortical tracts alone (column 1).

Figure 7 provides a further insight into these mappings, by plotting the connectivity patterns of the human regions against the pattern of its identified best match in the macaque brain. As can be observed, despite the relative proximity of these frontal regions, we have distinct patterns across them. With the exception of *dmPFC*, the connectivity patterns of all other regions have major contributions from the frontal striatal bundle, the extreme capsule and the amygdalofugal tract (*AMF*) and connection patterns to these cortico-subcortical bundles enable better separation of these nearby regions. For instance, *vmPFC* and *dmPFC* have both connections through the cingulum bundle, the corpus callosum and the inferior fronto-occipital fasciculus. However, they connect differently to the striatal bundle, the amygdalofugal tract and the anterior thalamic radiation and the addition of these tracts in the connectivity patterns allow the two regions to be better distinguished. The *FOp* and OFC_r have both relatively strong connection patterns to the uncinate and the inferior fronto-occipital fasciculi, but it is their different pattern of connections to extreme and external capsules and the amygdalofugal tract that enable their better separation.

3 Discussion

We introduced standardised dMRI tractography protocols for delineating cortico-subcortical connections between cortex and the amygdala, caudate, putamen and the hippocampus, across humans and macaques. Building upon our previous work (35, 31), which already provided protocols for cortico-thalamic radiations, and guided by the chemical tracer literature in the macaque, we devised the new protocols first for the macaque and then extended to humans. We demonstrated that our reconstructed tracts preserve topographical organisation principles, as suggested by tracers (62,8,63). As outlined in (43), tractography reconstructions can be highly accurate if information about where pathways go, and where they do not go is available. This is the philosophy behind the proposed protocols, which provide this type of constraints across

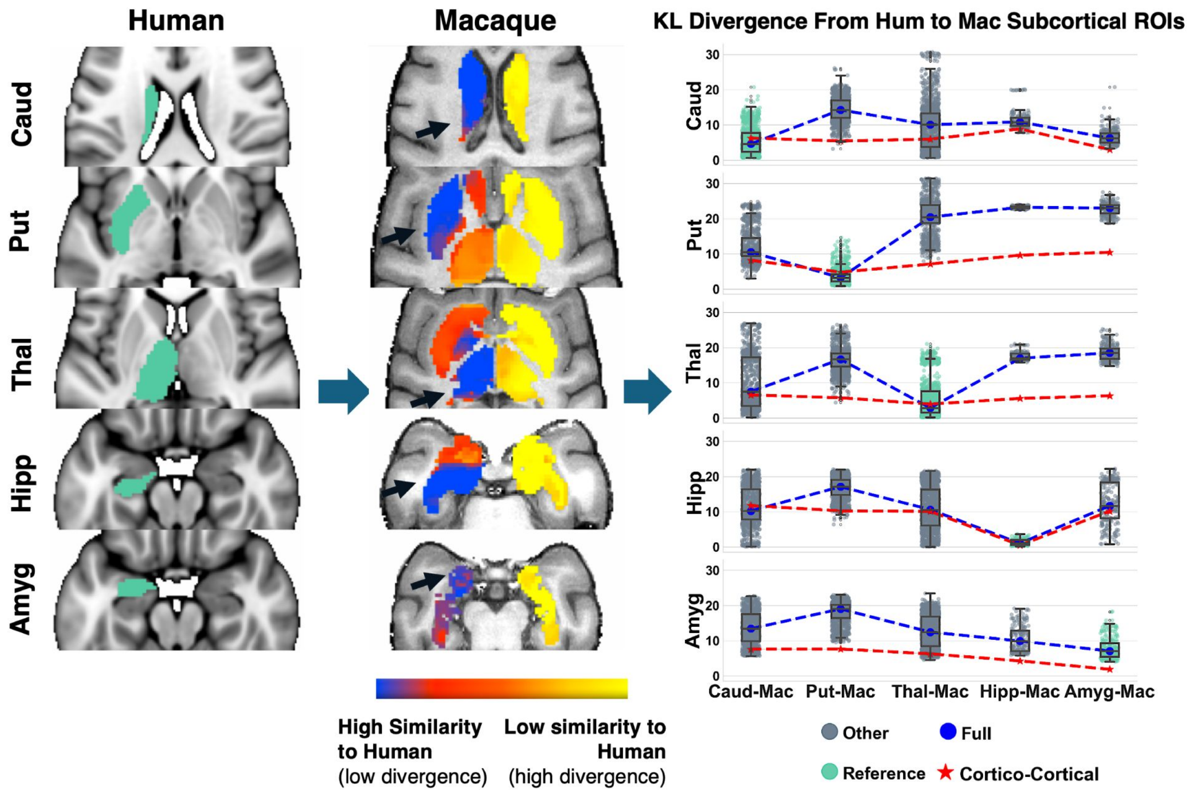

Figure 5

Identifying homologous deep brain structures (subcortical nuclei and hippocampus) across species solely by connectivity pattern similarity, obtained by the new tractography reconstructions.

Using the corresponding tracts in humans and macaques, connectivity blueprints can be calculated. These are GMxTracts matrices, with each row providing the pattern of how a GM location is connected to the predefined set of Tracts (35). **Left:** Starting from the average connectivity blueprints of reference human ROIs (Caud: Caudate, Put: Putamen, Thal: Thalamus, Hipp: Hippocampus, Amyg: Amygdala), KullbackLeibler (KL) divergence (or inverse similarity) maps can be computed against the connectivity blueprints of deeper subcortical regions in the macaque (**Middle**). The highest connection pattern similarity corresponds to the homologue macaque region of the corresponding human one. **Right**: Boxplots of KL divergence values between reference human regions and the five macaque ones. Blue dashed line corresponds to median KL divergence values when all white matter tracts are considered (both cortico-cortical and the new subcortical ones). Red dashed line corresponds to median KL divergence when using only cortico-cortical tracts. When cortico-subcortical tracts are included vs not, there is increased specificity/contrast in the cross-species mapping of these deeper structures. The boxplot with the lowest median divergence is shown in green in each case, indicating the best matching regions in the macaque to the human reference (i.e. caudate human reference best matches macaque caudate, putamen human reference best matches macaque putamen, etc).

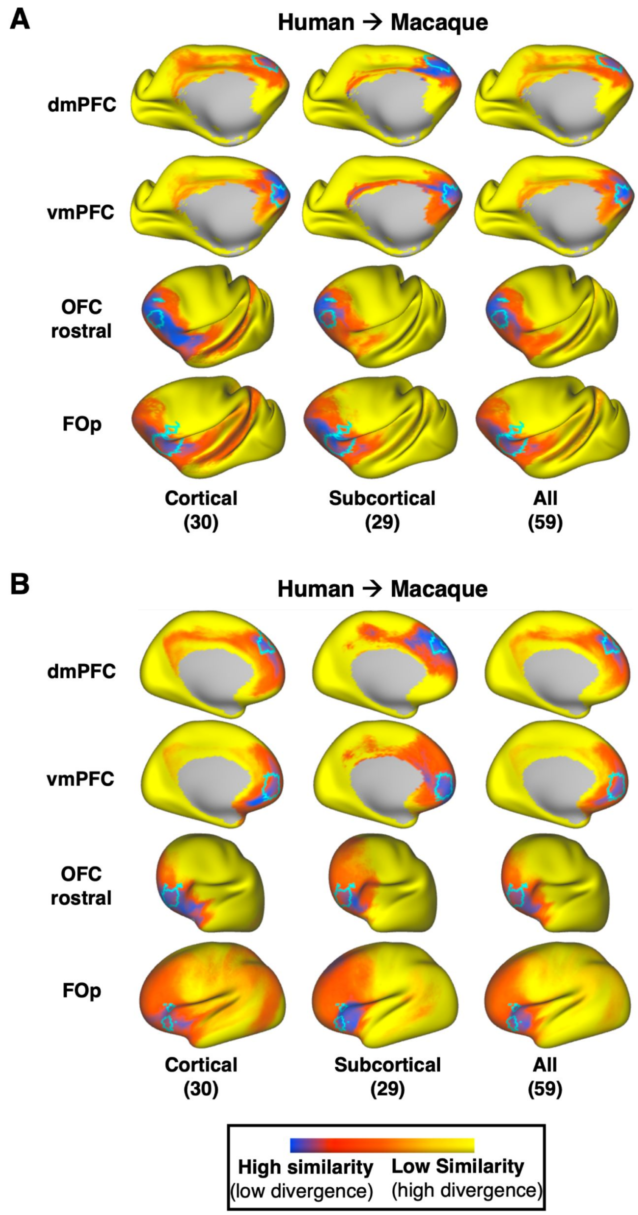

Figure 6

Identifying homologous cortical regions across species solely by connectivity pattern similarity, obtained with and without the new tractography reconstructions.

Two pairs of neighbouring frontal regions were chosen ($dmPFC$: dorsomedial prefrontal cortex & $vmPFC$: ventromedial prefrontal cortex, OFC_r: rostral orbitofrontal cortex & FOp: frontal operculum) and their mapping from human to macaque (A) and from macaque to human (B) was explored. For comparison, we overlay in cyan the corresponding homologue regions in each species, as defined in (26). (A) Kullback-Leibler (KL) divergence maps in the macaque for a given human cortical reference region, (one region per row), representing the similarity in connectivity patterns across the macaque cortex to the average pattern of the human reference region. KL divergence maps are calculated using cortico-cortical (1^{st} column), cortico-subcortical 2^{nd} column) and all tracts (3^{rd} column) to highlight the effect of the cortico-subcortical tractography reconstructions in the prediction. Subcortical tracts provide larger benefits for the prediction of $vmPFC$ and OFC_r, increasing specificity with respect to the expected borders. (B) Same as in A, but using macaque regions as reference and making predictions on the human cortex. KL divergence maps in the human for a given macaque cortical region, representing the similarity of connectivity pattern across the human cortex to the average pattern of the reference macaque region. Overall, in both species, an increased similarity to the reference regions in the homologue areas and decreased similarity across the rest of the cortex is observed, when cortico-subcortical tracts are considered (2^{nd} or 3^{rd} column).

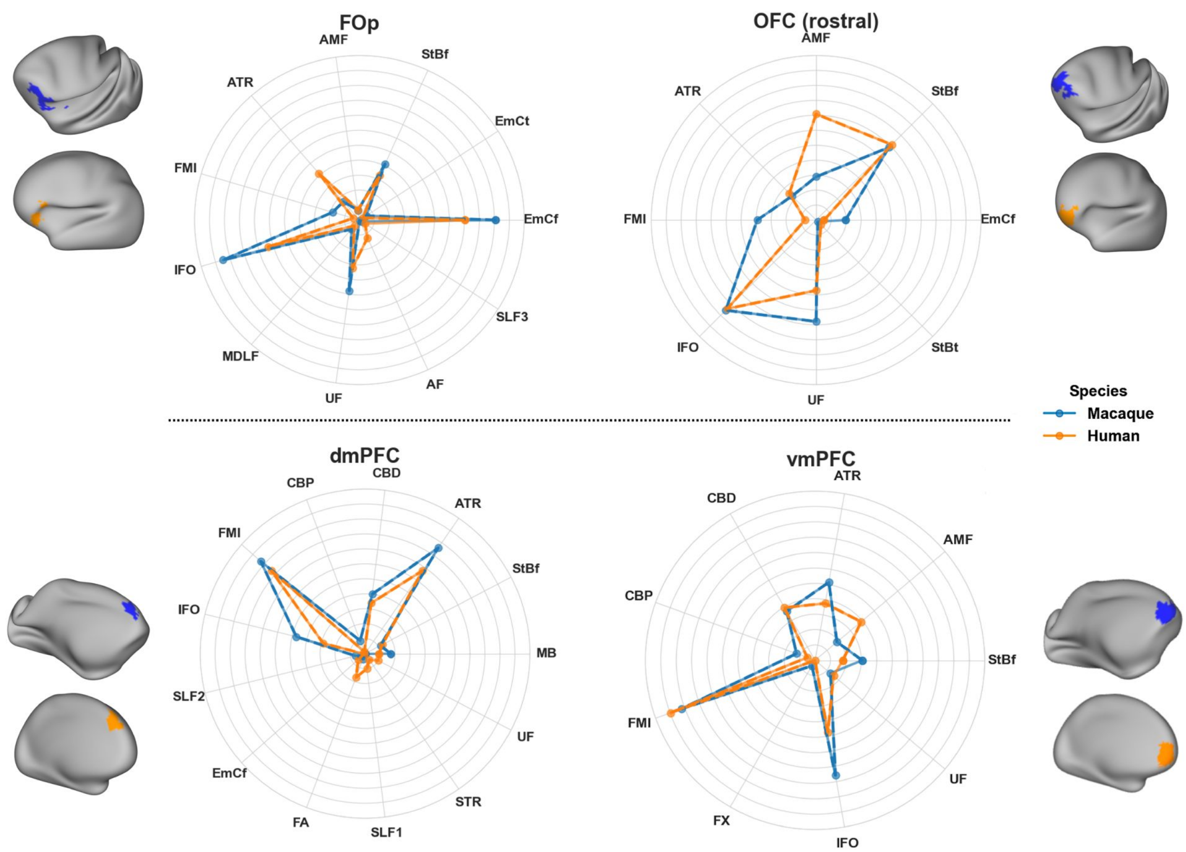

Figure 7

Connectivity patterns for neighbouring frontal region pairs, showing distinct cortico-subcortical tract contributions in macaque and human.

Considered regions are the same as in **Figure 6A**, i.e. FOp: frontal operculum, OFC_r: rostral orbitofrontal cortex, $dmPFC$: dorsomedial prefrontal cortex, $vmPFC$: ventromedial prefrontal cortex. Reference regions were chosen in the human cortex, shown in orange, and obtained from (26). The best matching region across the whole macaque cortex was identified by the minimum KL divergence in connectivity patterns (thresholded at the 7^{th} percentile in each case) and is shown in blue. Average connectivity patterns for the reference and best-matching regions are depicted using the polar plots. For each region, similarities in the connectivity patterns between the macaque and human can be observed, with the new cortico-subcortical bundles contributing to these patterns. For instance, FOp has a strong connection pattern involving EMC_f and UF and moderately StB_f and AF, while its neighbouring OFC_r has a stronger pattern involving StB_f and UF, compared to EMC_f. These differences are preserved across both species.

different bundles. At the same time these constraints are relatively coarse so that they are species-generalisable. We found that the proposed approaches yield tractography reconstruction across a range of datasets and respect individual similarities stemming from twinship. We further assessed the efficacy of these protocols in performing connectivity-based identification of homologous cortical and subcortical areas across the two species (35, 36, 38).

Mapping white matter tracts that link cortical areas with deep brain structures (subcortical nuclei and hippocampus), as done here, enhances capabilities for studying neuroanatomy in many contexts, from evolution and development, to mental health and neuropathology. As one of the (evolutionarily) older brain structures, the subcortex modulates brain functions including basic emotions, motivation, and movement control, providing a foundation upon which the more complex cognitive abilities of the cortex could develop and evolve (62,64,8,65,66). This modulatory function is mediated via white matter bundles (8, 9). Consequently, their disruption is linked to abnormal function and pathology, in mental health, neurodegenerative, and neurodevelopmental disorders (15, 67, 17). For example, in depression, fronto-thalamic (68), cortico-amygdalar (69, 70), and corticostriatal (71) connectivity changes have been reported, while in schizophrenia there are associated fronto-striatal (72) and hippocampal connectivity (73) changes. In Parkinson's there is impairment in fronto-striatal connectivity (74, 75, 76, 77), while fronto-thalamic and cingulate connectivity are impaired in Alzheimer's disease (75, 78). Connectivity between the frontal lobe and the amygdala, thalamus, and striatum, as well as cingulum connectivity are impaired in obsessive compulsive disorder (OCD), autism spectrum disorder (ASD), and attention deficit hyperactivity disorder (ADHD) (79, 69, 14, 80). Therefore, reconstructing connectivity of these deep brain structures (striatum, thalamus, amygdala, hippocampus) in a standardised manner, as enabled by our proposed tools, allows for further investigation into a wide range of disorders.

In addition, tractography of connections linking to/from deep brain structures has been used or proposed for guiding neuromodulation interventions, for example, deep brain stimulation (DBS) (81, 82) or repetitive transcranial magnetic stimulation (rTMS) (67). DBS can inherently target subcortical structures and connectivity of subcortical circuits can be used to identify efficacious stimulation targets (83, 18). rTMS on the other hand modulates subcortical function indirectly by targetting the structurally connected cortical areas. For example, dmPFC has been targetted to modulate the reward circuitry, in cases of anhedonia, negative symptoms in schizophrenia and major depression disorder (MDD) (84, 85, 86), while the vmPFC has been used as a target to modulate the prefrontal-striatal network (part of the limbic system) and regulate emotional arousal/anxiety (87, 88, 89). Our results show a good mapping across species of both these cortical regions with specificity in their connectional patterns. Additionally, the motor cortex has been used as target to modulate cortico-striatal connectivity in general anxiety disorder (90, 91). We thus anticipate that having a standardised set of tracts linking the striatum, the hippocampus, the amygdala and the thalamus (all potential sites for stimulation) to specific cortical areas can assist the planning of interventions.

Our cross-species approach naturally lends itself to the study of evolutionary diversity. A number of comparative studies have revealed differences and similarities when comparing brain connectivity between humans and non-human primates (92), including macaques (35, 38) and chimpanzees (93). Our work naturally extends these efforts and provides new tools for studying this diversity in deeper structures and subcortical nuclei. The ever increasing availability of comparative MRI data (34, 94) allows the definition of similar protocols in more species, such as the gibbon (33, 95) or the marmoset monkey, and even across geometrically diverse brains depicting different stages of neurodevelopment (e.g. neonates vs adults) enabling concurrent studies of phylogeny and ontogeny (38)).

Our protocols have been developed and tested using FSL-XTRACT, but, in principle, are not specific to FSL. We have not evaluated performance with other tools, but these standard space protocols could be translated into other tractography approaches. As described before, the protocols are recipes with anatomical constraints, including regions to which the corresponding white matter pathways connect and regions they do not, constructed with cross-species generalisability in mind. Caution may be needed, however, if applying such protocols for segmenting whole-brain tractograms, as these can induce more false positives than tractography reconstructions from smaller seed regions and may require stricter exclusions.

Despite the potential demonstrated in this work, our study has limitations. As this is the first endeavour of this scale to map cortico-subcortical connections in a standardised manner and across two species, it is not exhaustive. Tracts linking the cortex to the striatum were prioritised as they are of increased relevance in human development and disease. However, expanding to include more tracts targetting other structures would provide a more holistic view. Our protocols were developed in the adult human brain. Future work will translate them to the infant brain (expanding on previous work (38)) to interrogate cortico-subcortical connectivity across development. Tractography validation is a challenge, as is validation for any indirect and non-invasive imaging approach. We explored and demonstrated generalisability of the proposed protocols, both within and across species. We also showed how the imaging-based reconstructions follow topographical organisation principles suggested by tracers.

4 Methods and Materials

4.1 Tractography protocols

Guided by tract tracing and neuroanatomy literature, we devised tractography protocols for 18 subcortical bundles (nine bilateral - **Table 1**) using the XTRACT approach (31). We also revised protocols for three more bundles (2 bilateral, 1 commissural), compared to their original version (31). All protocols followed two principles: i) comprised of seed/stop/target/exclusion regions of interest (ROIs) defined in template space, so that they are standardised and generalisable, ii) ROIs defined equivalently between macaques and humans to enable tracking of corresponding bundles across species. The human protocols were defined in MNI152 space. The macaque protocols were defined in F99 and also in NMT space.

The tracts included the amygdalofugal tract (*AMF*), the uncinate fasciculus (*UF*), anterior commissure (*AC*), sensorimotor, temporal, parietal and frontal parts of the striatal bundle (*StB*) / external capsule (*EC*), Muratoff bundle (*MB*)/subcallosal fasciculus, as well as the extreme capsule (*EmC*) parts that run close to the putamen connecting the insula to the frontal, temporal and parietal cortices. All XTRACT tracts (Original, Revised and New) are summarised in Supplementary Table S1.

Detailed protocol definitions are presented below and summarised in **Figure 8** (for completeness, the previously published thalamic radiations from (31) are presented in Supplementary Materials). Briefly, the *AMF* and *UF* protocols are a standard-space generalisation of the individual subjectlevel protocols presented in (26). For the remaining protocols, we first devised them in the macaque guided by tract tracer literature. Specifically, the approach we took was to first identify anatomical constraints from neuroanatomy literature for each tract of interest independently, derive and test these protocols in the macaque. Thus, each devised protocol included a unique combination of anatomically defined masks (based on literature descriptions of the tracts), delineated in standard macaque space (F99). We then developed corresponding protocols in the human using correspondingly defined landmarks (delineated in standard MNI space). We optimised in an iterative fashion based on two criteria: (1) the protocols generalise well to humans, and (2) when considering groups of bundles, the generated reconstructions follow topographical principles known from tract tracing literature.

We modified existing XTRACT protocols to improve their specificity in the subcortex. Specifically, we developed a new uncinate fasciculus (*UF*) protocol based on the protocol presented in (26). We also modified the anterior commissure (*AC*) protocol to improve temporal lobe projections and slightly enhance projections to the amygdala, and modified the fornix (*FX*) one to reduce amygdala projections by placing an exclusion in the amygdala.

Given the proximity of the newly defined tracts, we evaluated the new protocols against their ability to capture patterns known from the tracer literature. These included relative positioning of each tract with respect to neighbouring tracts (**Figure 2**) and topographical organisation of certain bundle terminals within subcortical nuclei (**Figure 3**).

4.1.1 New protocol definitions

Amygdalofugal Pathway (*AMF*)

We derived generalisable template-space protocols to recon-struct the limbic-cortical ventral amygdalofugal pathway, following the subject-specific protocols in (26). The amygdalofugal pathway (*AMF*) courses between the amygdala and the prefrontal cortex (PFC), running alongside the uncinate fasciculus (*UF*) medially and finally merging with the *UF* in the posterior orbitofrontal cortex (OFC). As in (26), the seed included voxels with high fractional anisotropy in an anterior-posterior direction in the sub-commissural white matter. We used a target covering all brain at the level of caudal genu of the corpus callosum (same target as in the revised *UF* protocol, described further below). Exclusions include an axial plane through the *UF*, the internal and external capsules, the corpus callosum, the cingulate, the Sylvian fissure, the anterior commissure, the fornix, and a large coronal exclusion covering all brain dorsal to the corpus callosum and extending inferiorly to the frontal operculum and insula at the level of the middle frontal gyrus.

Striatal Bundle (StB_f, StB_m, StB_p, StB_t)

The striatal bundle (*StB*), is a bundle system that connects the cortex to the striatum, and joins the external capsule (*EC*). Although terminations reach both the caudate and the putamen, it primarily terminates in the putamen (48,49,10). Here we defined protocols for bundles that connect the putamen with frontal (including anterior cingulate) lobe, sensorimotor cortex, parietal and temporal lobes. For all parts, we used the putamen as the target. For StB_f, the OFC, PFC, ACC and frontal pole made up the seed. For StB_m, the primary sensorimotor cortex (M1-S1) were used as seeds. For StB_t and StB_p, the temporal and parietal lobes were respectively used as a seed. The exclusion masks shared many commonalities but also had differences. For all parts, exclusions included a midsagittal plane, the subcortex, except for the putamen, as well as the occipital lobe. For each of the parts, we additionally excluded the seeds for every other *StB* bundle.

Muratoff Bundle (*MB*)

The subcallosal fasciculus tract (as called in human neuroanatomy) or Muratoff bundle (as called in non-human animal neuroanatomy) (48, 51) is a complex system of projection fibres which runs beneath the corpus callosum, above the caudate nucleus at the corner formed by the internal capsule and the corpus callosum (42). Although terminations reach both the caudate and the putamen, it primarily terminates in the caudate head (48, 42, 51). As its cortical projections are challenging to capture and isolate using tractography, we defined a protocol for the major core of the bundle. We used a seed in the white matter adjacent to the caudate head and a target in the

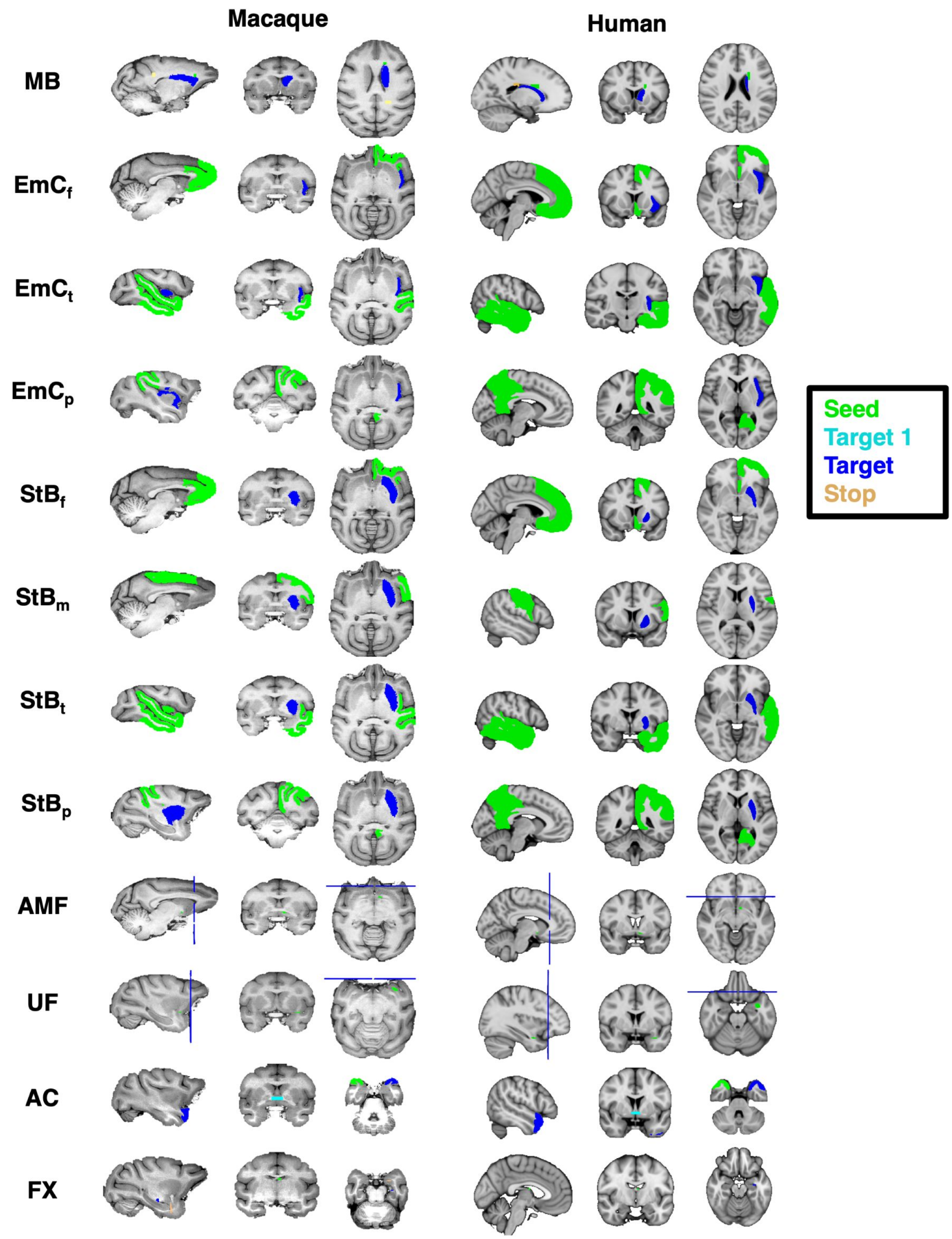

Figure 8

Corresponding tract protocol definitions across species.

Protocol definitions for all new (and revised) tracts in the human and macaque. Protocols were first designed in the macaque brain guided by macaque tracer literature, and then transferred over to the human. Colour-coded regions depict the seed, target and stop masks. Exclusion masks are not shown for ease of visualisation.

white matter adjacent to the caudate tail. A stop mask was used beyond the target in the white matter above the target. Exclusions included the contralateral hemisphere, the subcor-tex (except for the caudate head), the brainstem, the parietal, occipital, frontal and temporal cortices.

Extreme Capsule (EmC_f, EmC_p, EmC_t)

The extreme capsule is a major association fascicle that carries association fibres between frontal-temporal and frontal-parietal, as well as these areas and the insula (49). It lies between the claustrum and the insula, with the claustrum being considered the boundary between the *EmC* and the external capsule (*EC*) (10). We defined protocols connecting the insula to frontal, parietal and temporal cortices. For all parts we used the insula as the target, while for seeds we used the same seeds as for the corresponding *StB* parts. Hence, for the EmC_f protocol the frontal pole was used as the seed. For EmC_p, the parietal lobe was a seed, and for EmC_t the temporal lobe was a seed. Exclusions for all *EmC* parts included the contralateral part of the brain, the subcortex, as well as the occipital lobe, and lateral parts of the somatosensory and motor cortices. In addition, for each subdivision, the exclusion mask also included the seed mask for every other subdivision.

4.1.2 Revisions to previous XTRACT protocols

Uncinate Fasciculus (*UF*)

The *UF* lies at the bottom part of the extreme capsule, curving from the inferior frontal cortex to the anterior temporal cortex. Given the neighbouring bundles that were newly defined, we took a new approach to the *UF* compared to the original XTRACT implementation (31), now following the principles of (26). Briefly, we used an axial seed in the white matter rostro-laterally to the amygdala in the anterior temporal lobe. A target covered all brain at the level of caudal genu of the corpus callosum. Exclusions included the basal ganglia, a coronal plane posterior to the seed, the corpus callosum, the cingulate, the Sylvian fissure, the anterior commissure, and a large coronal exclusion covering all brain dorsal to the corpus callosum and extending inferiorly to the frontal operculum and insula at the level of the middle frontal gyrus. This implementation provided improved connectivity to the dorsal frontal cortex and aided separability with respect to neighbouring white matter bundles.

Anterior Commissure (*AC*)

Compared to original XTRACT protocol, we entirely re-worked the *AC* protocol. Previously, the mid-line main body of the *AC* was the seed with targets either side and stops at the amygdala. Now, we use a temporal pole as the seed, the main body of the *AC* as a waypoint and the contralateral temporal pole as the final target. For the human, we use the Harvard-Oxford temporal pole ROI (56). For the macaque, we use the CHARM temporal pole ROI (54). The temporal pole seed/target pair are flipped and tractography is repeated, taking the average of runs. Compared to the previous version, this protocol provides greater symmetry in resultant reconstructions and greater connectivity to the poles of the temporal cortex, as suggested in the literature (96, 97, 98, 10), and slightly enhanced connectivity to the amygdala.

Fornix (*FX*)

For the *FX*, a main output tract of the hippocampus, we have added an exclusion mask to the amygdala to prevent *FX* leakage to the amygdala, thus providing a cleaner *FX* compared to the original XTRACT implementation (31). For the human protocol, we used the Harvard-Oxford amygdala ROI (55). For the macaque, we used the SARM amygdala ROI (99).

4.2 Data

4.2.1 Macaque MRI data

We used six high quality ex-vivo rhesus macaque dMRI datasets, available from PRIME-DE (100). As described in (35, 31), these were acquired using a 7T Agilent DirectDrive console, with a 2D diffusion-weighted spin-echo protocol with single-line readout protocol with 16 volumes acquired at b=0 s/mm^2, 128 volumes acquired at b=4000 s/mm^2, and a 0.6 mm isotropic spatial resolution.

4.2.2 Human MRI data

We used high quality minimally preprocessed (101) in-vivo dMRI data from the young adult Human Connectome Project (HCP) (57, 28). The HCP data were acquired using a bespoke 3T Connectom Skyra (Siemens, Erlangen) with a monopolar diffusion-weighted (Stejskal-Tanner) spin-echo echo planar imaging (EPI) sequence with an isotropic spatial resolution of 1.25 mm, three shells (b values = 1000, 2000, and 3000 s/mm^2), and 90 unique diffusion directions per shell plus 6 b = 0 s/mm^2 volumes, acquired twice with opposing phase encoding polarities. Data correspond to total scan time per subject of approximately 55 minutes. For this study, we randomly drew 50 HCP subjects (age range, 22-36 years of age, 24/26 females/males.

To assess robustness against data quality, we also used data from the UK Biobank (3T Prisma, 32 channel coil, 2 mm isotropic resolution, b values = 1000 and 2000 s/mm^2, 50 directions per shell). The UK Biobank data are acquired with approximately 6.5 minutes scan time per subject and therefore represent more standard quality datasets, achievable in a clinical scanner (58). Fifty subjects were randomly drawn from the UK Biobank (UKB) (age range 42-65 years of age, 31/19 females/males). For both HCP and UKB cohorts we ensured that the distribution of QC metrics (such as subject motion and image SNR/CNR) was representative of the full HCP and UKB cohorts that we had available.

4.2.3 Macaque tracer data

Tracer data were used to test aspects of the striatal bundle protocols (**Figure 3**). These were made available by SRH and were obtained from an existing collection of injections in 19 macaque brains, from (102, 103, 104, 105, 106, 62, 107, 48, 108, 50, 109) and cases from the laboratory of SRH. Specifically, anterograde tracers were injected across 78 cortical locations and their terminations within the putamen were recorded in coronal slices of the NMT template space at 0.5 mm resolution. Specifically, the injection sites were first assigned to one of four cortical ROIs (frontal, parietal, temporal and sensorimotor cortices), obtained from the NMT CHARM v1 parcellation (54). For each of these four injection ROIs, we counted all the corresponding terminations within the putamen, and then divided by the total number of termination sites. This resulted in a termination probability map for each cortical region across the putamen, and these termination maps were smoothed using spline interpolation. The putamen mask was obtained from the NMT SARM v1 parcellation (110). To compare against tractography in F99 space, these maps were non-linearly registered from NMT to F99 space using RheMAP (111).

4.3 MRI data preprocessing

4.3.1 Crossing fibre modelling and tractography

For both the human and macaque data, we modelled fibre orientations for up to three orientations per voxel using FSL's BEDPOSTX (112, 113). These orientations were used in tractography. Probabilistic tractography was performed using FSL's XTRACT (31), which uses FSL's PROB-TRACKX (114,115). The standard space protocol masks were used to seed and guide tractography, which occurred in diffusion space for each dataset. 60 major white matter fibre bundles were reconstructed (30 cortico-cortical, 29 cortico-subcortical, 1 cerebellar, Supplementary Table S1). A curvature threshold of 80° was used, the maximum number of streamline steps was 2000, and subsidiary fibres were considered above a volume fraction threshold of 1%. A step size of 0.5 mm was used for the human brain, and a step size of 0.2 mm was used for the macaque brain. Resultant spatial path distributions were normalised by the total number of valid streamlines.

4.3.2 Registration to standard space

For the human data, nonlinear transformations of T1-weighted (T1w) to MNI152 standard space were obtained. The distortion-corrected dMRI data were separately linearly aligned to the T1w space, and the concatenation of the diffusion-to-T1w and T1w-to-MNI transforms allowed diffusion-to-MNI warp fields to be obtained. For the macaque, nonlinear transformations to the macaque F99 standard space were estimated using FSL's FNIRT (116) based on the corresponding FA maps.

For cases where NMT-space tractography protocols were used, nonlinear transformations to NMT space were obtained using RheMAP (111).

4.4 Tractography against data quality and individual variability

4.4.1 Varying data quality

To explore robustness against varying data quality, we compared tractography reconstructions for in-vivo human dMRI data of considerably different data resolutions, diffusion contrast and scan time. Specifically, we explored whether tract reconstructions in state-of-the-art HCP data (approximately 55 minutes of scan time) were similar to reconstructions in bog standard data from the UK Biobank (UKB) (approximately 6.5 minutes of scan time), both on group average maps, as well as individual reconstructions.

Inter-subject variability for each tract reconstruction was assessed within and across the HCP and UKB cohorts. Inter-subject Pearson's correlations were obtained by cross correlating random subject pairs tract-wise. Specifically, for each subject pair, we correlated the normalised path distributions in MNI space for each tract, after thresholding the path distribution at 0.5% (31), and then averaged the correlation across tracts for each subject pair. This was repeated for all possible unique subject pairs within and across cohorts.

A pairwise Mann-Whitney U-test was performed to determine differences in variability across analyses. For example we compared the HCP vs UKB correlation between original and the new(+revised) tracts. We corrected for multiple comparisons using Bonferroni correction.

We also explored tract reconstructions on individual subjects. To demonstrate representative results, we ranked subjects based on their tractography results against the cohort average and picked the the 10^{th}, 50^{th} (median) and 90^{th} percentile of the subjects. Specifically, for each subject, we calculated the average Pearson's correlation value, to the group average, across all tracts. We then ranked the subjects based on this value.

4.4.2 Respecting similarities stemming from twinship

As an indirect way to explore whether the proposed standardised protocols respected individual variability, we tested whether tractography reconstructions reflected similarities stemming from twinship. We used the family structure in the HCP cohort, to explore whether tracts of monozygotic twin pairs were more similar compared to tract similarity in dizygotic twins and non-twin sibling pairs, and to tract similarity in unrelated subject pairs, as would be expected by heritability of structural connections (59,117,60). We used the 72 pairs of monozygotic twins (MZ) available in the HCP cohort, and randomly selected 72 pairs of dizygotic twins (DZ), 72 pairs of non-twin siblings and 72 pairs of unrelated subjects, to have a balanced comparison. We compared tracts across pairs to assess whether our automated protocols respect the underlying tract variability across individuals. Specifically, for a given subject pair and a given tract, we calculated the Pearson's correlation between the normalised path distributions (in MNI space and following thresholding at 0.5%). We repeated for all tracts and then calculated the mean correlation and standard deviation across tracts for that subject pair. This was then repeated for each group of subject pairs giving a distribution of average correlations for each group. We subsequently compared these distributions between the different groups. We repeated this process separately for the Original XTRACT tracts (31) and the new cortico-subcortical tracts to ensure that patterns were similar. For each analysis, a pairwise Mann-Whitney U-test was performed for all cohort pairs to determine the significant differences between them. We corrected for multiple comparisons using Bonferroni.

4.5 Building connectivity blueprints in cortex and subcortex

Connectivity blueprints are $GM \times Tracts$ matrices that have been proposed to represent the pattern of connections of GM areas to a predefined set of WM tracts (35, 36). To do so, the intersection of the core of WM tracts with the WM-GM boundary needs to be identified. For cortical GM, simply obtaining the intersection from the spatial path distribution maps of each tract would be dominated by the gyral bias in tractography near the cortex (118). Instead, whole-brain tractography matrices can be used as intermediaries. Specifically a $GM \times WM$ connectivity matrix can be generated by seeding from each location of the white-grey matter boundary and targetting to a whole WM mask and this can then be multiplied by a $WM \times Tracts$ obtained by collating the path distributions of all tracts of interest.

The cortical blueprints $GM_{ctx} \times Tracts$ were generated using our previously developed tool xtract blueprint (35, 38). We used the GM-WM boundary surface, extracted using the HCP pipelines (101) for the human data and the approach in (35) for the macaque. Briefly, a single set of macaque surfaces were derived using a set of high-quality structural data from one of the macaque subjects. The remaining macaque data were then nonlinearly transformed to this space, and the surfaces were nonlinearly transformed to the F99 standard space. All surface data were downsampled to 10,000 vertices prior to tractography. Volume space white matter targets were downsampled to 3 mm isotropic for the human and 2 mm isotropic for the macaque.

We extended the blueprint generation to include the subcortex. For subcortical nuclei we found that using an intermediary $GM \times WM$ matrix did not help (as gyral bias is not relevant in subcortex - in fact it made patterns less specific). Hence, subcortical $GM_{sub} \times Tracts$ blueprints were built using the intersection of the path distribution of each tract with the subcortical structures of interest (i.e. through multiplication of white matter tracts and binary subcortical masks, including putamen, caudate, thalamus, hippocampus, amygdala). **Figure 9**) shows a comparison of the two approaches for various tracts in the human and macaque: i) using an intermediary $GM \times WM$ matrix to obtain subcortical connection patterns, as done in (35) for cortical regions and ii) using directly the tractography path distributions. The latter approach gave more focal and specific patterns and was used here for the subcortical regions. Tracts were downsampled (at 2 mm for human and 1 mm for macaque), thresholded at 0.1%, and multiplied by the subcortical nuclei masks, and then vectorised and stacked to create a $GM_{sub} \times Tracts$ matrix. These were then row-wise concatenated (i.e vertically) with the cortical blueprints to generate CIFTI-style blueprints with approximately 10,000 cortical vertices and approximately 5,000 subcortical voxels (per left/right hemisphere). Finally, connectivity blueprints were row-wise sum-normalised. Following subjectwise construction of connectivity blueprints, we derived group-averaged blueprints for macaques and humans.

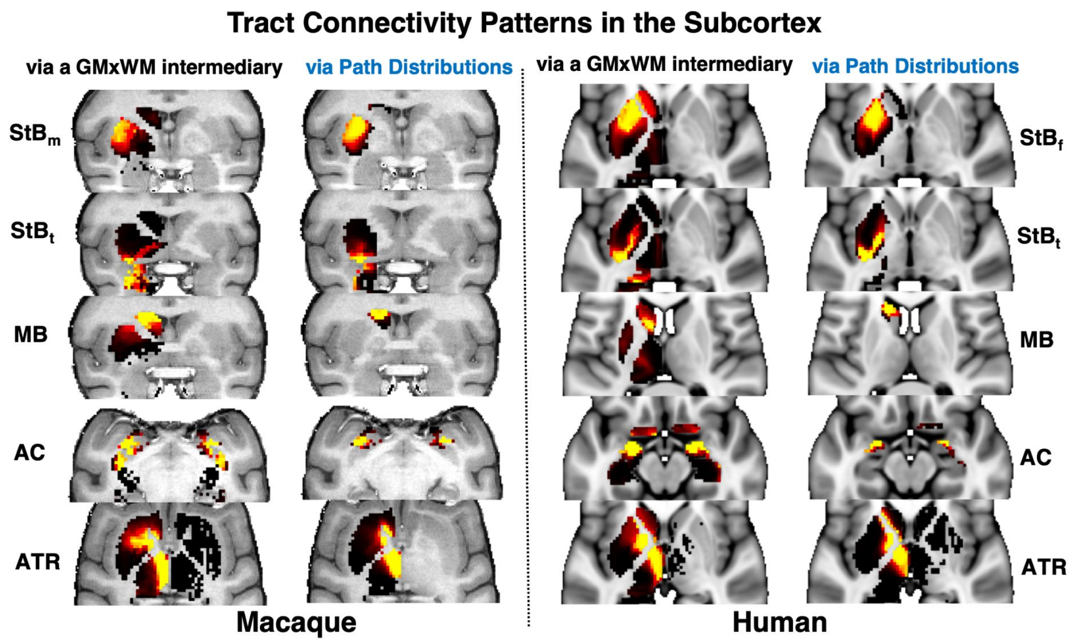

Figure 9

Improved specificity in subcortical connectivity patterns when using directly the tractography path distributions.

Subcortical $GM_{sub} \times Tracts$ blueprints were built using: i) an intermediary whole brain tractography $GM \times WM$ matrix, multiplied by $WM \times Tracts$ as done in (35) for cortical regions, ii) the intersection of the path distribution of each tract with the subcortical structures of interest. The two approaches are shown on the left and right columns for each of the macaque and human examples and for representative example tracts (rows). The latter approach resulted in improved specificity in both the macaque and human, with the tract of interest connecting more focally to the relevant subcortical nucleus. For instance StB tracts end up more specifically in the putamen, MB in the caudate, AC in the amygdala and ATR in the thalamus. All examples are shown as axial views, apart from StB_m, StB_t, MB in the macaque that are shown in coronal views.

4.6 Comparing connectivity blueprints across species

We compared grey matter connectivity patterns between humans and macaques (i.e. rows of the corresponding connectivity blueprint matrices), both in cortex and subcortex. As connectivity patterns are anchored by sets of homologously defined white matter landmarks, connectivity patterns may be compared statistically using Kullback-Leibler (KL) divergence (Equation 1) (119), as previously used (35).

Let M be the macaque connectivity blueprint matrix, with M_{ik} linking grey matter (cortex or subcortex) location i to tract k = 1: T, with the set of tracts with length T. Let matrix H be the equivalent matrix for the human brain. Vertices i and j in the macaque and human brains can then be compared in terms of their connectivity patterns M_{ik}, $H_{j\,k}$, k = 1: T using the symmetric KL divergence $D_{i\,j}$ as a dissimilarity measure. To avoid degeneracies in KL divergence calculations induced by the presence of zeros, we shifted all blueprint values by $\delta = 10^{-6}$. We used the tool xtract divergence to perform all relevant calculations.

$$D_{ij} = \sum_k M_{ik} \log_2 \frac{M_{ik}}{H_{jk}} + \sum_k H_{jk} \log_2 \frac{H_{jk}}{M_{ik}} \quad (1)$$

Supplementary Materials

Original cortico-thalamic XTRACT protocols

For completeness of presenting all the cortico-subcortical protocols, we include here the previously published protocols of thalamic radiations (31).

Acoustic Radiation (*AR*)

The acoustic radiation connects the medial geniculate nucleus (*MGN*) of the thalamus to the auditory cortex. The seed was placed in the transverse temporal gyrus and the target covered the *MGN* of the thalamus. The exclusion mask consisted of two coronal planes, anterior and posterior to the thalamus, and an axial plane superior to the thalamus. In addition, the exclusion mask contained the brainstem and a horizontal region covering the optic tract.

Anterior Thalamic Radiation (*AT R*)

The anterior thalamic radiation connects the thalamus to the frontal lobe. The seed mask was a coronal mask through the anterior part of the thalamus (120), with a coronal target mask at the anterior thalamic peduncle. In addition, the exclusion mask contained an axial plane covering the base of the midbrain, a coronal plane preventing leakage via the posterior thalamic peduncle and a coronal plane preventing leakage via the cingulum. A coronal stop mask covered the posterior part of the thalamus, extending from the base of the midbrain to the callosal sulcus.

Optic Radiation (*OR*)

The optic radiation consists of fibres from the lateral geniculate nucleus (*LGN*) of the thalamus to the primary visual cortex. The seed was placed in the *LGN* and the target mask consisted of a coronal plane through the anterior part of the calcarine fissure. Exclusion masks consisted of an axial block of the brainstem, a coronal block of fibres directly posterior to the *LGN* to select fibres that curl around dorsally, and a coronal plane anterior to the seed to prevent leaking into longitudinal fibres.

Superior Thalamic Radiation (*ST R*)

The superior thalamic radiation connects the thalamus to the pre-/post-central gyrus respectively. The seed was a mask covering the whole thalamus and the target an axial plane covering the superior thalamic peduncle. An axial plane was used as a stop mask ventrally to the thalamus. The exclusion mask included two coronal planes, anterior and posterior to the target, to exclude tracking to the prefrontal cortex and occipital cortex respectively.

Data and materials availability

Human in-vivo diffusion MRI data are publicly available (*www.humanconnectome.org/*) and provided by the Human Connectome Project (HCP), WU-Minn Consortium (Principal Investigators: David Van Essen and Kamil Ugurbil; 1U54MH091657) funded by the 16 NIH Institutes and Centres that support the NIH Blueprint for Neuroscience Research; and by the McDonnell Centre for Systems Neuroscience at Washington University (57). The UK Biobank data were used under UK Biobank Project 43822 (PI: Sotiropoulos). Macaque data are openly available (*https://fcon_1000.projects.nitrc.org/indi/PRIME/oxford2.html*) and provided via the PRIMatE Data Exchange (*http://fcon_1000.projects.nitrc.org/indi/PRIME/oxford2.html*) (100).

Tractography protocols (*https://github.com/SPMIC-UoN/xtract_data*) and white matter tract atlases (*https://github.com/SPMIC-UoN/XTRACT_atlases*) will be made available on GitHub and will be released in FSL. Tools for performing standardised and automated tractography (XTRACT), building connectivity blueprints (xtract blueprint) and performing divergencebased comparisons of connectivity blueprints (xtract divergence) are available on GitHub (*https://github.com/SPMIC-UoN/xtract*) and are released in FSL (v6.0.7.10 onwards, https://fsl.fmrib.ox.ac.uk/fsl/docs/#/diffusion/xtract).

Additional information

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Author contributions

All authors contributed significantly to this work from different and, in cases, overlapping roles. The specific contributions of each are: SA: Conceptualisation, Methodology, Data Curation, Formal Analysis, Investigation, Visualisation, Writing – Original Draft, Writing – Review and Editing SW: Methodology, Data Curation, Investigation, Writing – Review and Editing DF: Methodology, Formal Analysis, Writing - Review and Editing KB: Resources, Methodology, Writing - Review and Editing AMK: Resources, Methodology WT: Resources, Methodology, Writing - Review and Editing SJ: Methodology, Resources, Writing – Review and Editing SH: Resources, Methodology, Writing – Review and Editing RM: Methodology, Formal Analysis, Writing – Review and Editing SNS: Conceptualisation, Methodology, Formal Analysis, Resources, Writing – Original Draft, Writing – Review and Editing, Supervision, Funding Acquisition

Cortico-Cortical	Abbreviation	Bilateral	Version
Arcuate Fasciculus	AF	Yes	Original
Cingulum subsection : Dorsal	CBD	Yes	Original
Cingulum subsection : Peri-genual	CBP	Yes	Original
Cingulum subsection : Temporal	CBT	Yes	Original
Corticospinal Tract	CST	Yes	Original
Frontal Aslant	FA	Yes	Original
Forceps Major	FMA	No	Original
Forceps Minor	FMI	No	Original
Inferior Longitudinal Fasciculus	ILF	Yes	Original
Inferior Fronto-Occipital Fasciculus	IFO	Yes	Original
Middle Longitudinal Fasciculus	$MdLF$	Yes	Original
Superior Longitudinal Fasciculus 1	$SLF1$	Yes	Original
Superior Longitudinal Fasciculus 2	$SLF2$	Yes	Original
Superior Longitudinal Fasciculus 3	$SLF3$	Yes	Original
Vertical Occipital Fasciculus	VOF	Yes	Original
Uncinate Fasciculus	UF	Yes	Revised
Cortico-Subcortical			
Acoustic Radiation	AR	Yes	Original
Anterior Thalamic Radiation	ATR	Yes	Original
Optic Radiation	OR	Yes	Original
Superior Thalamic Radiation	STR	Yes	Original
Fornix	FX	Yes	Revised
Anterior Commissure	AC	No	Revised
Amygdalofugal Tract	AMF	Yes	New
Muratoff Bundle/Subcallosal Fasciculus	MB	Yes	New
Striatal Bundle/External Capsule (sensorimotor)	StB_m	Yes	New
Striatal Bundle/External Capsule (frontal)	StB_f	Yes	New
Striatal Bundle/External Capsule (temporal)	StB_t	Yes	New
Striatal Bundle/External Capsule (parietal)	StB_p	Yes	New
Extreme Capsule (frontal)	EmC_f	Yes	New
Extreme Capsule (temporal)	EmC_t	Yes	New
Extreme Capsule (parietal)	EmC_p	Yes	New
Cerebellar			
Middle Cerebellar Peduncle	MCP	No	Original

Table S1

List of all species-matched (human & macaque) tract protocols

, grouped as cortico-cortical, cortico-subcortical and cerebellar. Columns indicate whether corresponding protocols are bilateral or not and whether they are New, Revised or not changed (Original XTRACT).

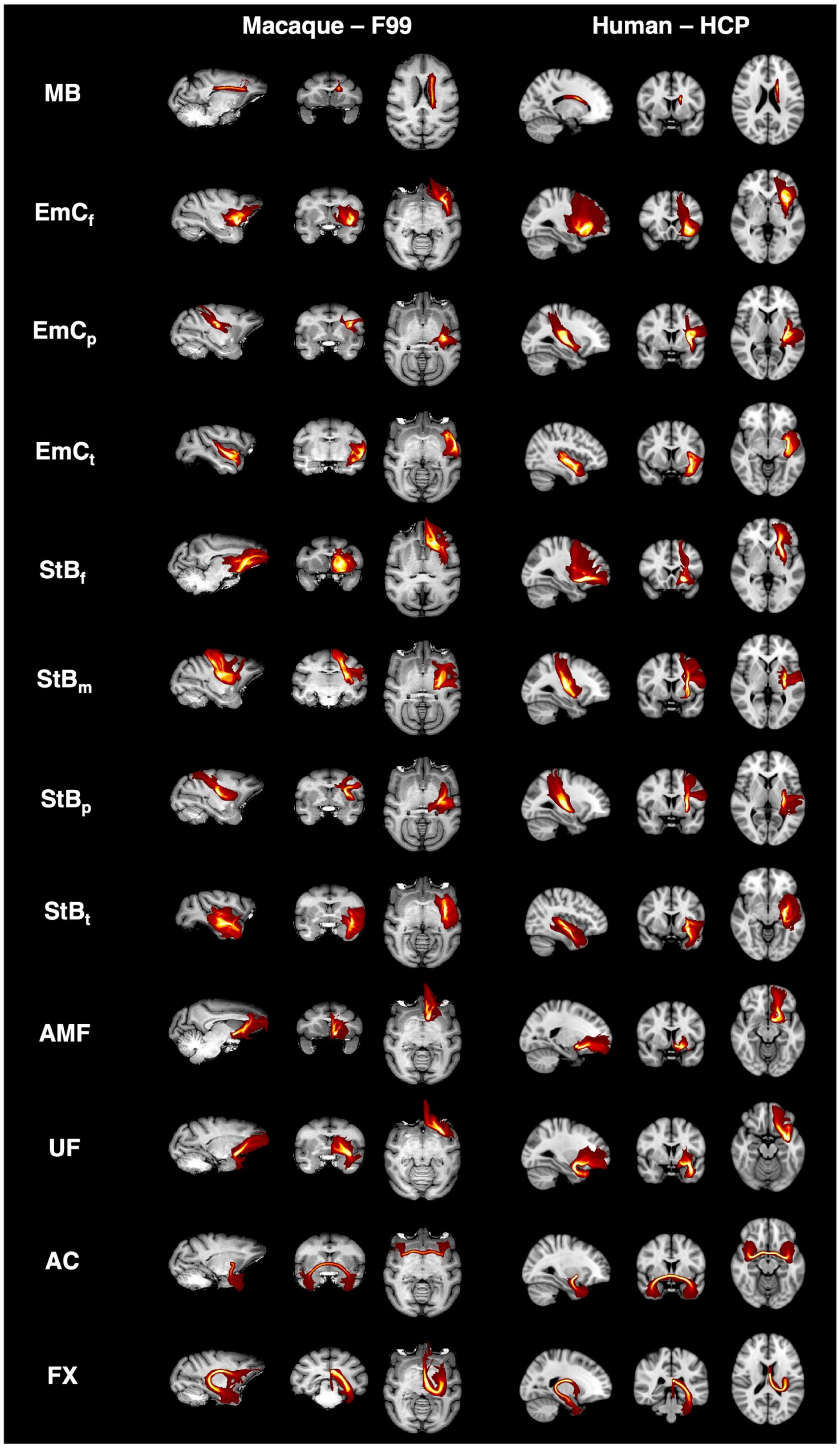


Figure S1

Tract reconstructions using our cortico-subcortical protocols, with good agreement between the macaque and the human.

Maximum intensity projections (MIPs) of the group-averaged path distributions for all developed tractography protocols in the macaque (6 animal average) and human (50 healthy subject average from the Human Connectome Project dataset; HCP). All MIPs are across a window (20% of the field of view) centred at the displayed slices. Thresholded path distributions are displayed with a low threshold of 0.1% (for the EmC parts the 90^{th} percentile was used).

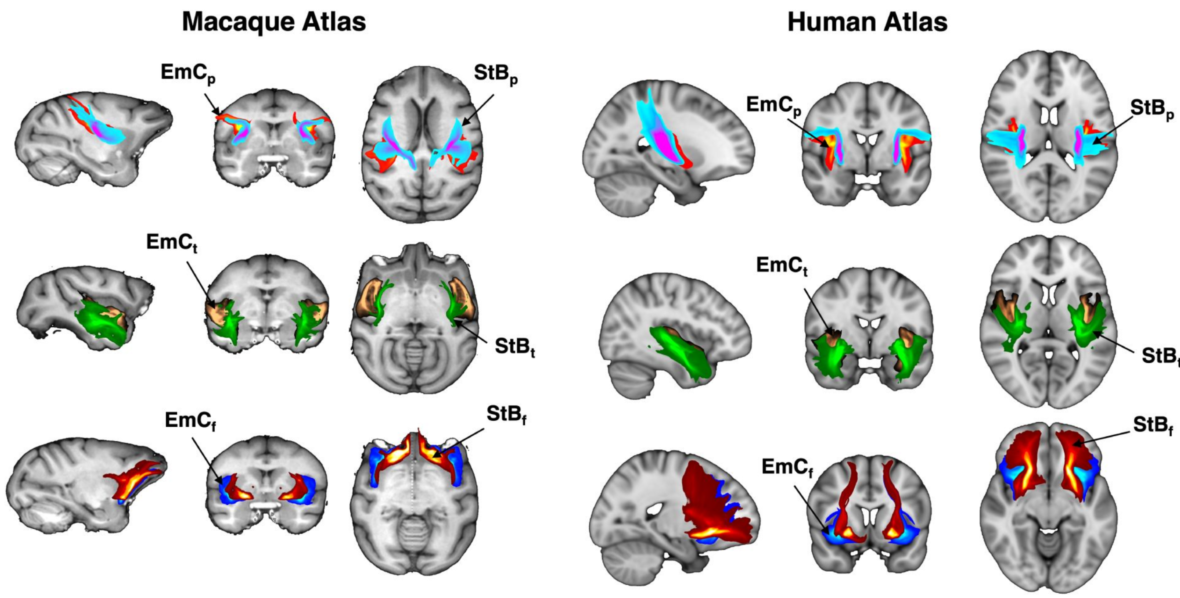


Figure S2

Relative positions maintained for all corresponding parts of striatal and extreme capsule bundles, across species.

Maximum intensity projections (MIPs) of the group-averaged tractography results for corresponding parts of the striatal bundle (StB)/external capsule and the extreme capsule (EmC) in the macaque (6 animal average) and human (50 healthy subject average from the Human Connectome Project dataset; HCP). For each part, StB is more medial and EmC is more lateral (with respect to each other. Tracts considered: frontal, temporal and parietal parts of the anterior limb of the extreme capsule (EmC_f, EmC_t, EmC_p)*; frontal, temporal and parietal parts of the striatal bundle (*StB_f, StB_t, StB_p*) (***Table 1*** in main text)*.

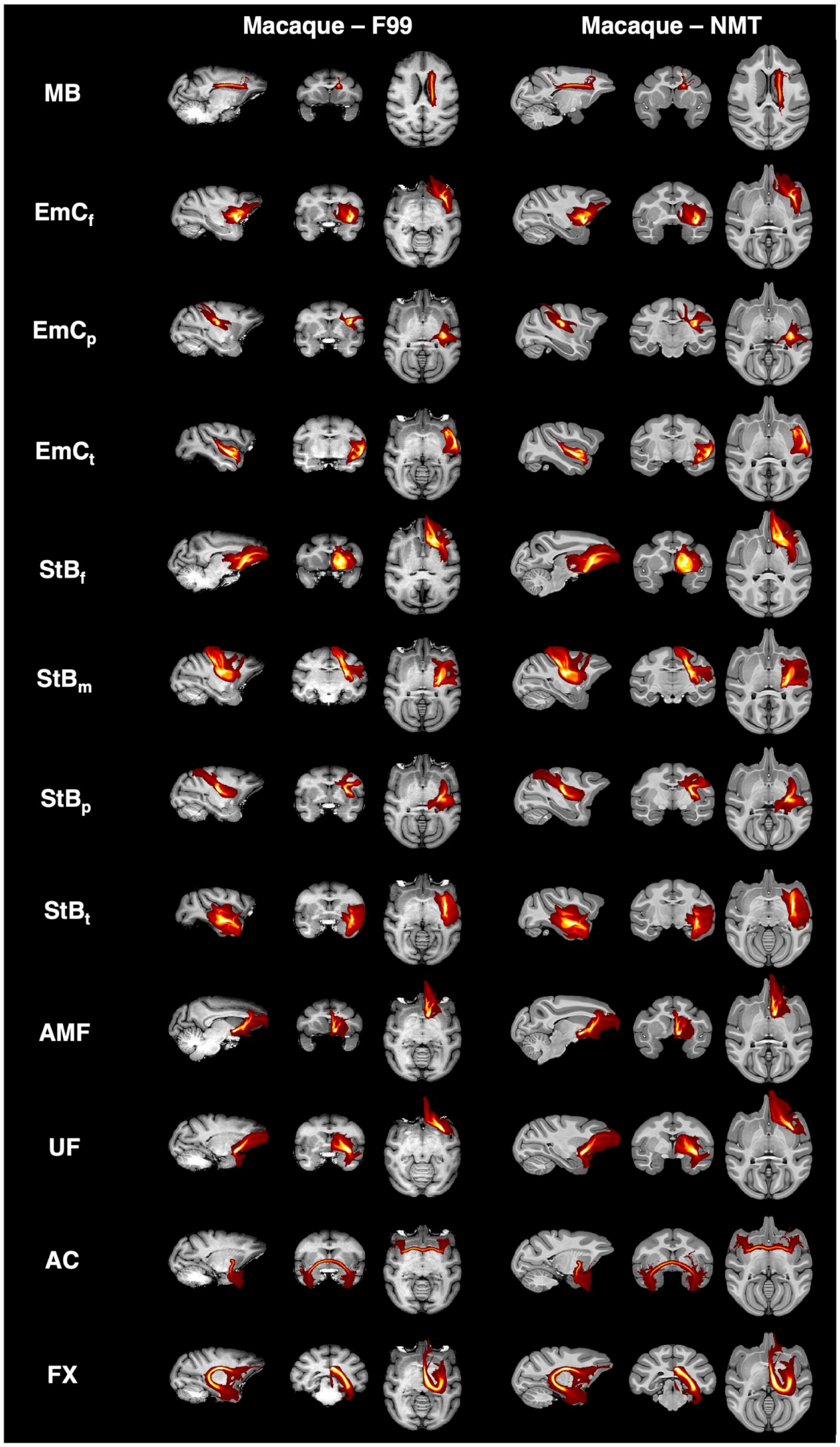


Figure S3

Corresponding tract reconstructions using our macaque cortico-subcortical protocols, between the two macaque standard spaces, F99 and NMT.

Maximum intensity projections (MIPs) of the groupaveraged path distributions for all developed tractography results for all developed protocols in the macaque (6 animal average) using protocols in the F99 standard space and protocols in the NMT standard space. All MIPs are across a window (20% of the field of view) centred at the displayed slices. Thresholded path distributions are displayed with a low threshold of 0.1% (for the EmC parts the 90th percentile was used).

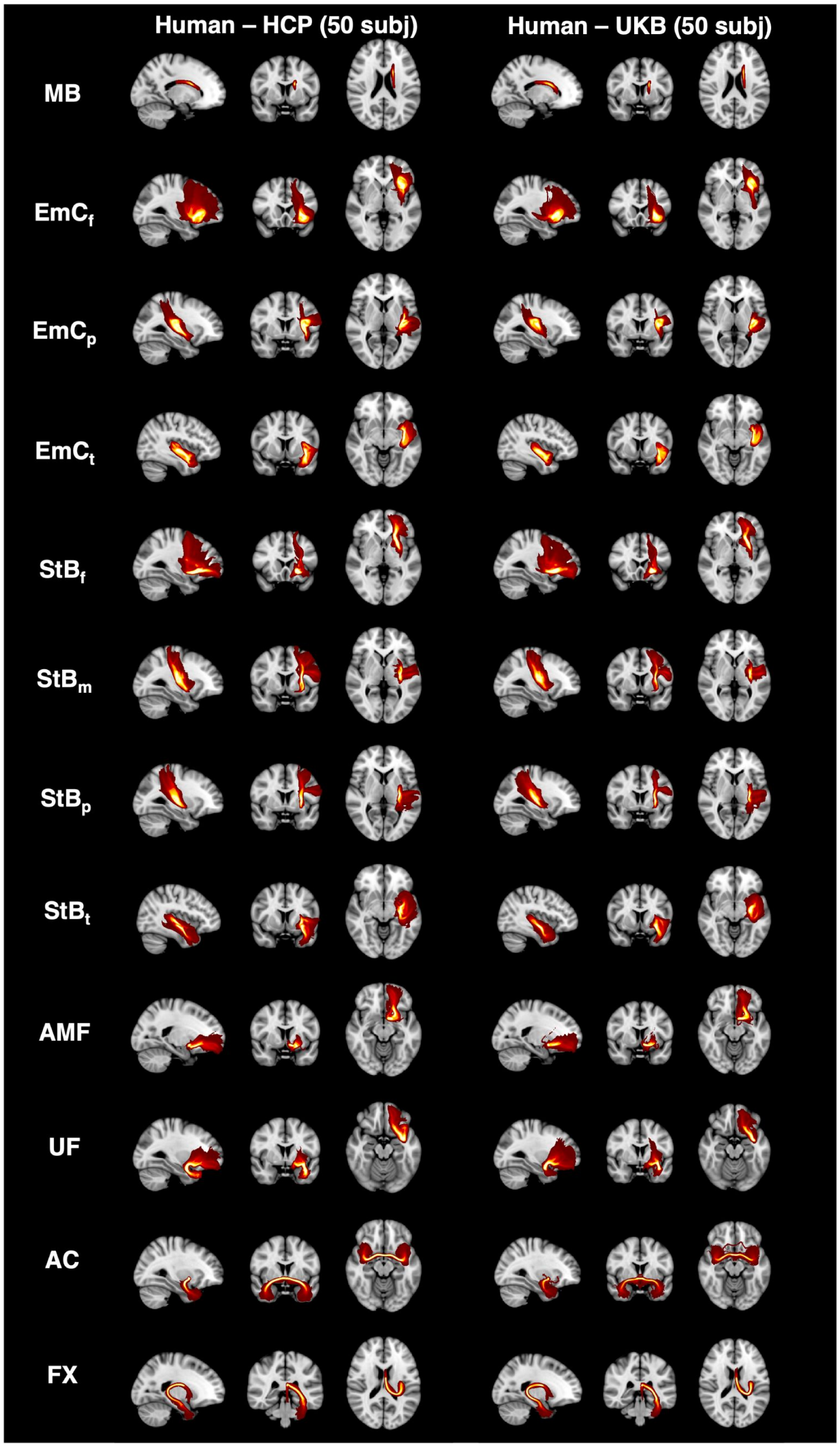


Figure S4

Corresponding tract reconstructions using our cortico-subcortical protocols, between different data resolutions (spatial and angular) and acquisition protocols.

Maximum intensity projections (MIPs) of the group-averaged path distributions for all developed tractography results for all developed protocols in 50 Human Connectome Project (HCP) and 50 UK Biobank (UKB) subjects. All MIPs are across a window (20% of the field of view) centred at the displayed slices. Thresholded path distributions are displayed with a low threshold of 0.1% (for the EmC parts the 90^{th} percentile was used).

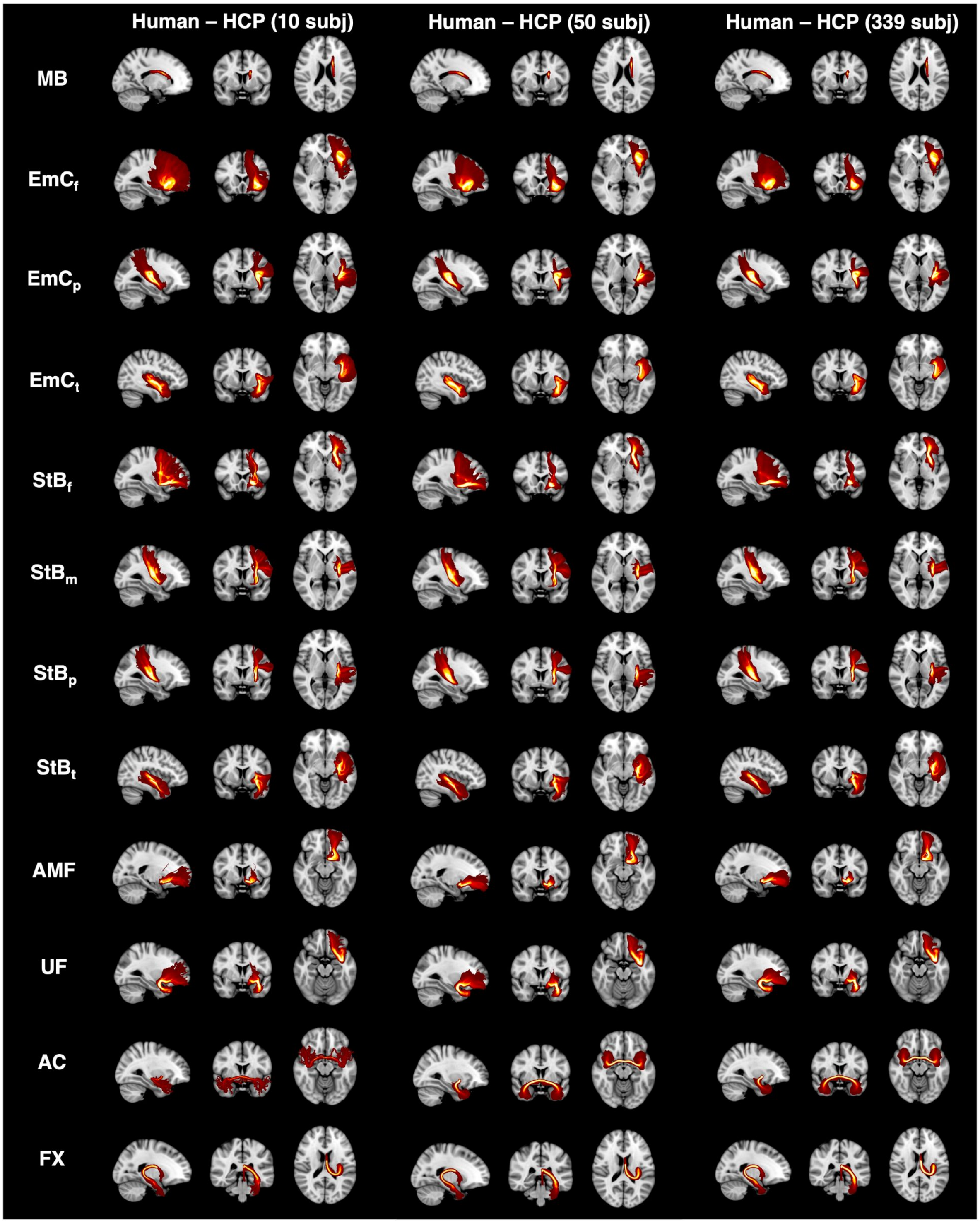


Figure S5

Corresponding tract reconstructions using our cortico-subcortical protocols, across different cohort sizes.

Maximum intensity projections (MIPs) of the group-averaged path distributions for all developed tractography results for all developed protocols in 10, 50 and 339 (all unrelated subjects) Human Connectome Project (HCP) subjects. All MIPs are across a window (20% of the field of view) centred at the displayed slices. Thresholded path distributions are displayed with a low threshold of 0.1% (for the EmC parts the 90^{th} percentile was used).

Figure S6

Individual subject tract reconstructions match the group average, whilst preserving expected topology for

StB **and** EmC. Showing reconstruction and relative topology for frontal and parietal parts (StB_f vs EmC_f, StB_p vs EmC_p) for individual subjects. The subjects chosen are those corresponding to the 10^{th}, 50^{th} (median) and 90^{th} percentile of the distribution of tract correlations against the group average for the HCP cohort (N=50). For each subject, the mean correlation to the average across all New tracts was computed and subjects were ranked based on this mean tract correlation. In all cases, tracts reconstruct similarly to the average atlas, whilst preserving the expected topology (StB more medial than EmC).

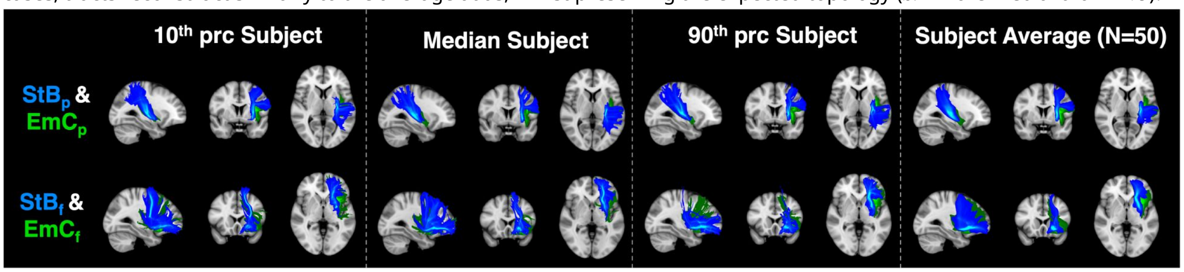
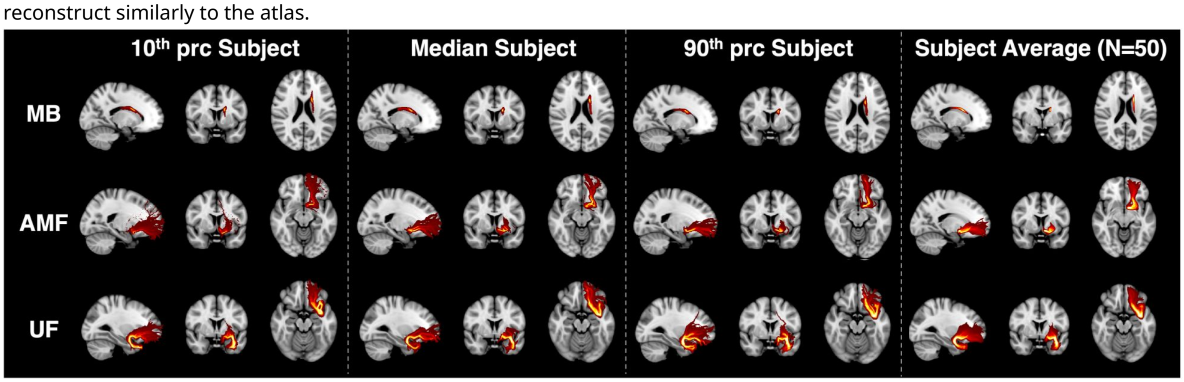


Figure S7

Individual subject tract reconstructions match the group average for

MB, AMF **and** UF. The subjects chosen are those corresponding to the 10^{th}, 50^{th} (median) and 90^{th} percentile of the tract correlations against the group average for the HCP cohort (N=50). For each subject, the mean correlation to the average across all New tracts was computed and subjects were ranked based on this mean tract correlation.. In all cases tracts reconstruct similarly to the atlas.



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Reviewer #1 (Public review):

The authors note that it is challenging to perform diffusion MRI tractography consistently in both humans and macaques, particularly when deep subcortical structures are involved. The scientific advance described in this paper is effectively an update to the tracts that the XTRACT software supports. The changes to XTRACT are soundly motivated in theory (based on anatomical tracer studies) and practice (changes in seeding/masking for tractography).

https://doi.org/10.7554/eLife.107012.2.sa2

Reviewer #2 (Public review):

Summary:

In this article, Assimopoulos et al. expand the FSL-XTRACT software to include new protocols for identifying cortical-subcortical tracts with diffusion MRI, with a focus on tracts connecting to the amygdala and striatum. They show that the amygdalofugal pathway and divisions of the striatal bundle/external capsule can be successfully reconstructed in both macaques and humans while preserving large-scale topographic features previously defined in tract tracing studies. The authors set out to create an automated subcortical tractography protocol, and they accomplish this for a subset of specific subcortical connections.

Strengths:

The main strength of the current study is the translation of established anatomical knowledge to a tractography protocol for delineating cortical-subcortical tracts that are difficult to reconstruct. Diffusion MRI-based tractography is highly prone to false positives; thus, constraining tractography outputs by known anatomical priors is important. The authors used existing tracing literature to create anatomical constraints for tracking specific cortical-subcortical connections and refined their protocol through an iterative process and in collaboration with multiple neuroanatomists. Key additional strengths include 1) the creation of a protocol that can be applied to both macaque and human data; 2) demonstration that the protocol can be applied to be high quality data (3 shells, > 250 directions, 1.25 mm isotropic, 55 minutes) and lower quality data (2 shells, 100 directions, 2 mm isotropic, 6.5

minutes); and 3) validation that the anatomy of cortical-subcortical tracts derived from the new method are more similar in monozygotic twins than in siblings and unrelated individuals.

Overall Appraisal:

This new method will accelerate research on anatomically validated cortical-subcortical white matter pathways. The work has utility for diffusion MRI researchers across fields.

Editors' note:

Both reviewers were satisfied with the responses to their feedback.

https://doi.org/10.7554/eLife.107012.2.sa1

Author response:

The following is the authors' response to the original reviews.

> ***Reviewer #1 (Public Review):***
>
> *(1) Summary:*
>
> *The authors note that it is challenging to perform diffusion MRI tractography consistently in both humans and macaques, particularly when deep subcortical structures are involved. The scientific advance described in this paper is effectively an update to the tracts that the XTRACT software supports. The claims of robustness are based on a very small selection of subjects from a very atypical dMRI acquisition (n=50 from HCP-Adult) and an even smaller selection of subjects from a more typical study (n=10 from ON-Harmony).*
>
> *Strengths:*
>
> *The changes to XTRACT are soundly motivated in theory (based on anatomical tracer studies) and practice (changes in seeding/masking for tractography), and I think the value added by these changes to XTRACT should be shared with the field. While other bundle segmentation software typically includes these types of changes in release notes, I think papers are more appropriate.*

We would like to thank the reviewer for their assessment and we appreciate the comments for improving our manuscript. We have added new results, sampling from a larger cohort with a typical dMRI protocol (N=50 from UK Biobank), as well as showcasing examples from individual subject reconstructions (Supplementary figures S6, S7). We also demonstrate comparisons against another approach that has been proposed for extracting parts of the cortico-striatal bundle in a bundle segmentation fashion, as the reviewer suggests (see comment and Author response image 1 below).

We would also like to take the opportunity to summarise the novelty of our contribuIons, as detailed in the Introduction, which we believe extend beyond a mere software update; this is a byproduct of this work rather than the aim.

i) We devise for the first Ime standard-space protocols for 21 challenging cortico-subcortical bundles for both human and macaque and we interrogate them in a comprehensive manner.

ii) We demonstrate robustness of these protocols using criteria grounded on neuroanatomy, showing that tractography reconstructions follow topographical principles known from tracers both in WM and GM and for both species. We also show that these protocols capture individual variability as assessed by respecting family structure in data from the HCP twins.

iii) We use high-resolution dMRI data (HCP and post-mortem macaque) to showcase feasibility of these reconstructions, and we show that reconstructions are also plausible with more conventional data, such as the ones from the UK Biobank.

iv) We further showcase robustness and the value of cross-species mapping by using these tractography reconstructions to predict known homologous grey matter (GM) regions across the two species, both in cortex and subcortex, on the basis of similarity of grey matter areal connection patterns to the set of proposed white matter bundles.

> *Weaknesses*
>
> *(2) The demonstration of the new tracts does not include a large number of carefully selected scans and is only compared to the prior methods in XTRACT. The small n and limited statistical comparisons are insufficient to claim that they are better than an alternative. Qualitatively, this method looks sound.*

We appreciate the suggestion for larger sample size, so we performed the same analysis using 50 randomly drawn UK Biobank subjects, instead of ON-Harmony, matching the N=50 randomly drawn HCP subjects (detailed explanation in the comment below, Main text Figure 4A; Supplementary Figures S4). We also generated results using the full set of N=339 HCP unrelated subjects (Supplementary Figure S5 compares 10, 50 and 339 unrelated HCP subjects). We provide further details in the relevant point (3) below.

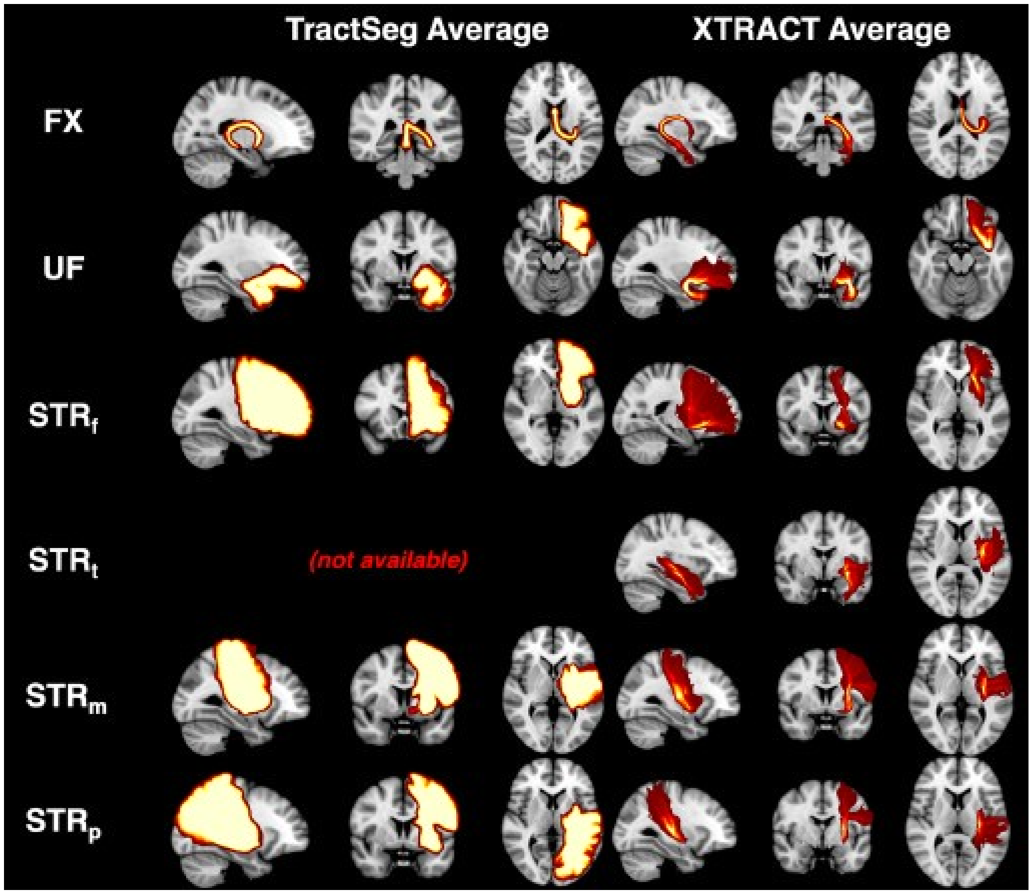
With regards to comparisons to other methods, there are not really many analogous approaches that we can compare against. In our knowledge there are no previous cross-species, standard space tractography protocols for the tracts we considered in this study (including Muratoff, amygdalofugal, different parts of extreme an external capsules, along with their neighbouring tracts). We therefore i) directly compared against independent neuroanatomical knowledge and patterns (Figures 2, 3, 5), ii) confirmed that patterns against data quality and individual variability that the new tracts demonstrate are similar to patterns observed for the more established cortical tracts (Figure 4), iii) indirectly assessed efficacy by performing a demanding task, such as homologue identification on the basis of the tracts we reconstruct (Figures 6, 7).

We need to point out that our approach is not “bundle segmentation”, in the sense of “datadriven” approaches that cluster streamlines into bundles following full-brain tractography. The latter is different in spirit and assigns a label to each generated streamline; as full-brain tractography is challenging (Maier-Hein, Nature Comms 2017), we follow instead the approach of imposing anatomical constraints to miIgate for some of these challenges as suggested in (MaierHein, 2017).

Nevertheless, we used TractSeg (one of the few alternatives that considers corticostriatal bundles) to perform some comparisons. The Author response image below shows average path distributions across 10 HCP subjects for a few bundles that we also reconstruct in our paper (no temporal part of striatal bundle is generated by Tractseg). We can observe that the output for each tract is highly overlapping across subjects, indicating that there is not much individual variability captured. We also see the reduced specificity in the connectivity end-points of the bundles.

Author response image 1.

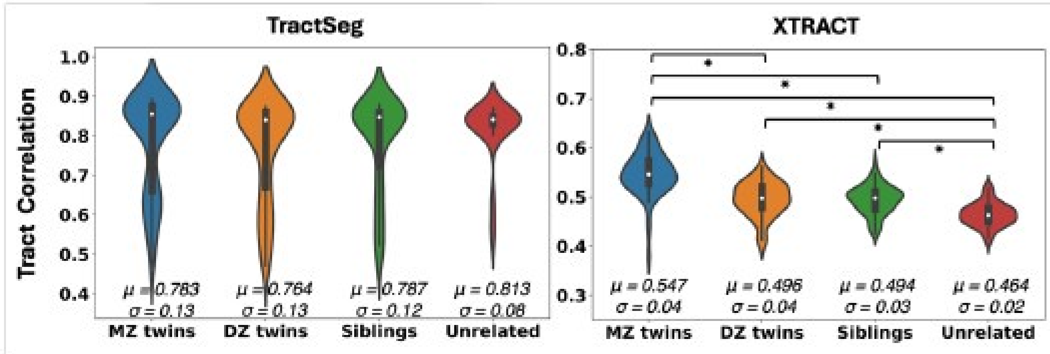
Comparison between 10-subject average for example subcortical tracts using TractSeg and XTRACT. We chose example bundles shared between our set and TractSeg. Per subject TractSeg produces a binary mask rather than a path distribution per tract. Furthermore, the mask is highly overlapping across subjects. Where direct correspondence was not possible, we found the closest matching tract. Specifically, we used ST_PREF for STBf, and merged ST_PREC with ST_POSTC to match StBm. There was no correspondence for the temporal part of StB.



We subsequently performed the twinness test using both TractSeg and XTRACT (Author response image 2), as a way to assess whether aspects of individual variability can be captured. Due to heritability of brain organisation features, we anticipate that monozygotic twins have more similar tract reconstructions compared to dizygoIc twins and subsequently non-twin siblings. This pattern is reproduced using our proposed approach, but not using TractSeg that provides a rather flat pattern.

Author response image 2.

Violin plots of the mean pairwise Pearson's correlations across tracts between 72 monozygotic (MZ) twin pairs, 72 dizygotic (DZ) twin pairs, 72 non-twin sibling pairs, and 72 unrelated subject pairs from the Human Connectome Project, using Tractseg (left) and XTRACT (right). About 12 cortico-subcortical tracts were considered, as closely matched as possible between the two approaches. For Tractseg we considered: 'CA', 'FX', 'ST_FO', 'ST_M1S1' (merged 'ST_PREC' and 'ST_POSTC' to approximate the sensorimotor part of our striatal bundle), 'ST_OCC', 'ST_PAR', 'ST_PREF', 'ST_PREM', 'T_M1S1' (merged 'T_PREC' and 'T_POSTC' to approximate the sensorimotor part of our striatal bundle), 'T_PREF', 'T_PREM', 'UF'. For XTRACT we considered: 'ac', 'fx', 'StB_f', 'StB_m', 'StB_p', 'StB_t, 'EmC_f', 'EmC_p', 'EmC_t', 'MB', 'amf', 'uf'. Showing the mean (μ) and standard deviation (σ) for each group. There were no significant di^erences between groups using TractSeg.



Taken together, these results indicate as a minimum that the different approaches have potentially different aims. Their different behaviour across the two approaches can be desirable and beneficial for different applications (for instance WM ROI segmentation vs connectivity analysis) but makes it challenging to perform like-to-like comparisons.

> *(3) "Subject selection at each stage is unclear in this manuscript. On page 5 the data are described as "Using dMRI data from the macaque (N = 6) and human brain (N = 50)". Were the 50 HCP subjects selected to cover a range of noise levels or subject head motion? Figure 4 describes 72 pairs for each of monozygotic, dizygotic, non-twin siblings, and unrelated pairs - are these treated separately? Similarly, NH had 10 subjects, but each was scanned 5 times. How was this represented in the sample construction?"*

We appreciate the suggestions and we agree that some of the choices in terms of group sizes may have been confusing. Short answer is we did not perform any subject selection, subjects were randomly drawn from what we had available. The 72 twin pairs are simply the maximum number of monozygotic twin pairs available in the HCP cohort, so we used 72 pairs in all categories to match this number in these specific tests. The N=6 animals are good quality post-mortem dMRI data that have been acquired in the past and we cannot easily expand. For the rest of the points, we have now made the following changes:

We have replaced our comparison to the ON-Harmony dataset (10 subjects) with a comparison to 50 unrelated UK Biobank subjects (to match the 50 unrelated HCP subject cohort used throughout). Updated results can be seen in Figure 4A and Supplementary Figure S4. This allows a comparison of tractography reconstruction between high quality and more conventional quality data for the same N.

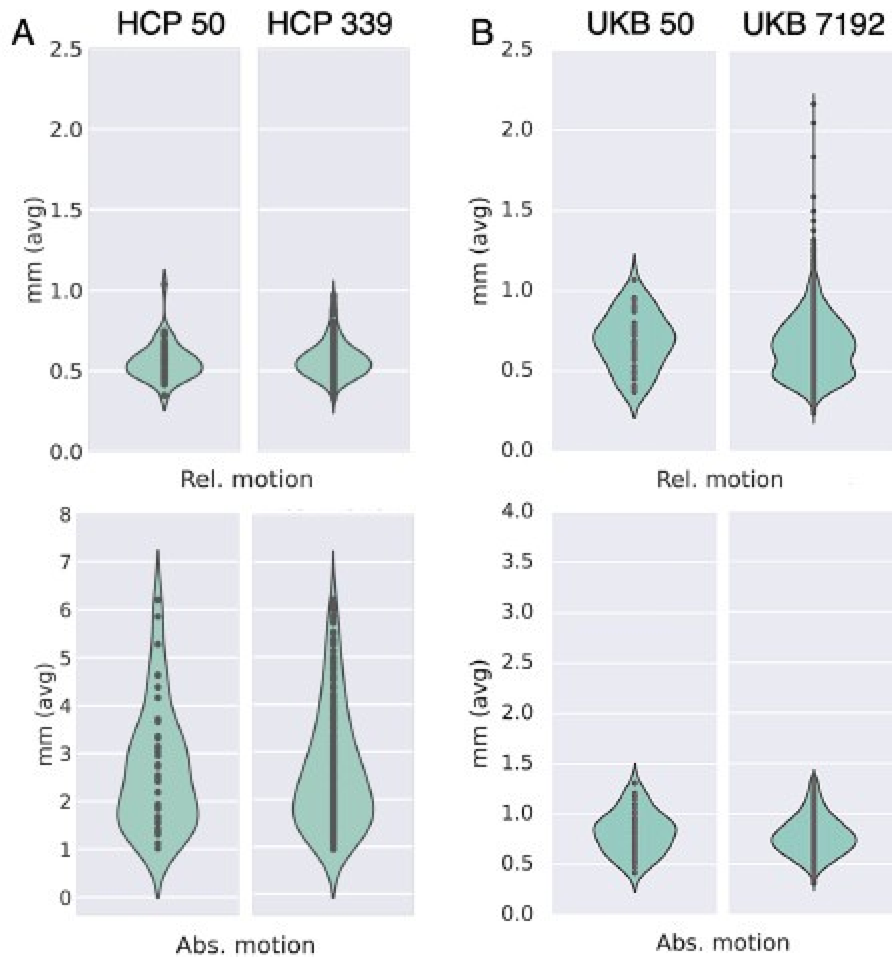
We looked at QC metrics to ensure our chosen cohorts were representaIve of the full cohorts we had available. The N=50 unrelated HCP cohort and N=50 unrelated UKBiobank cohorts we used in the study captured well the range of the full 339 unrelated HCP cohort and N=7192 UKBiobank cohort in terms of absolute/relative moion (Author response image 3A and 3B respectively). A similar pattern was observed in terms of SNR and CNR ranges Author response image 4).

We generated tractography reconstructions for single subjects, corresponding to the 10th percentile (P_{10}), median and 90th percentile (P90) of the distributions with respect to similarity to the cohort average maps. These are now shown in Supplementary Figures S6, S7. We also checked the QC metrics for these single subjects and confirmed that average absolute subject moIon was highest for the P_{10}, followed by the P_{50} and lowest for the P_{90} subject, capturing a range of within cohort data quality.

We generated reconstructions for an even larger HCP cohort (all 339 unrelated HCP subjects) and these look very similar to the N=50 reconstructions (Supplementary Figure S5).

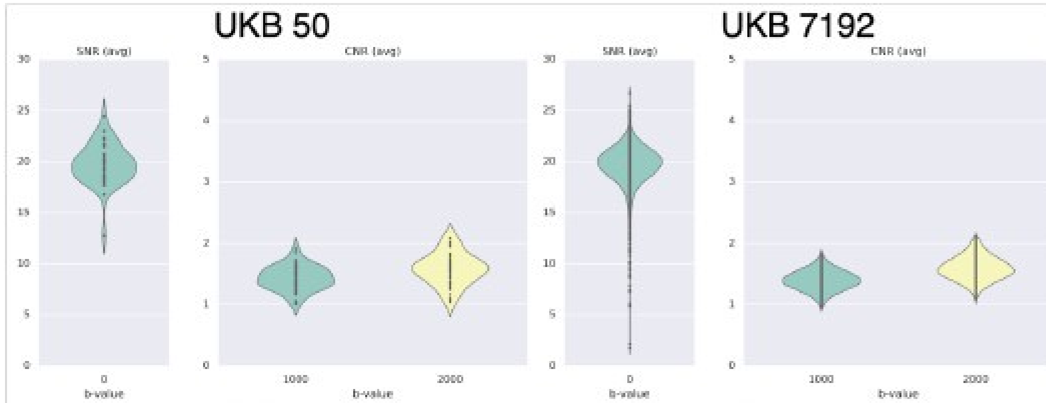
Author response image 3.

Subsets chosen from the HCP and UKB reflect similar range of average motion (relative and absolute) to the corresponding full cohorts. (A) Absolute and relative motion comparison between N=50 and N=339 unrelated HCP subjects. (B) Absolute and relative motion comparison between N=50 and N=7192 super-healthy UKB subjects.



Author response image 4.

Average SNR and CNR values show similar range between the N=50 UKB subset and the full UK Biobank cohort of N=7192.



> *(4) In the paper, the authors state "the mean agreement between HCP and NH reconstructions was lower for the new tracts, compared to the original protocols (p < 10^−10). This was due to occasionally reconstructing a sparser path distribution, i.e., slightly higher false negative rate," - how can we know this is a false negative rate without knowing the ground truth?*

We are sorry for the terminology, we have corrected this, as it was confusing. Indeed, we cannot call it false negaIve, what we meant is that reconstructions from lower resolution data for these bundles ended up being in general sparser than the ones from the high-resolution data, potentially missing parts of the tract. We have now revised the text accordingly.

> ***Reviewer #2 Public Review:***
>
> *(5) Summary:*
>
> *In this article, Assimopoulos et al. expand the FSL-XTRACT software to include new protocols for identifying cortical-subcortical tracts with diffusion MRI, with a focus on tracts connecting to the amygdala and striatum. They show that the amygdalofugal pathway and divisions of the striatal bundle/external capsule can be successfully reconstructed in both macaques and humans while preserving large-scale topographic features previously defined in tract tracing studies. The authors set out to create an automated subcortical tractography protocol, and they accomplished this for a subset of specific subcortical connections for users of the FSL ecosystem.*
>
> *Strengths:*
>
> *A main strength of the current study is the translation of established anatomical knowledge to a tractography protocol for delineating cortical-subcortical tracts that are difficult to reconstruct. Diffusion MRI-based tractography is highly prone to false positives; thus, constraining tractography outputs by known anatomical priors is important. Key additional strengths include 1) the creation of a protocol that can be applied to both macaque and human data; 2) demonstration that the protocol can be applied to be high quality data (3 shells, > 250 directions, 1.25 mm isotropic, 55 minutes) and lower quality data (2 shells, 100 directions, 2 mm isotropic, 6.5 minutes); and 3) validation that the anatomy of cortical-subcortical tracts derived from the new method are more similar in monozygotic twins than in siblings and unrelated individuals.*

We thank the Reviewer for the globally posiIve evaluaIon of this work and the perInent comments that have helped us to improve the paper.

> *Weaknesses*

> *(6) Although this work validates the general organizational location and topographic organization of tractography-derived cortical-subcortical tracts against prior tract tracing studies (a clear strength), the validation is purely visual and thus only qualitative. Furthermore, it is difficult to assess how the current XTRACT method may compare to currently available tractography approaches to delineating similar cortical-subcortical connections. Finally, it appears that the cortical-subcortical tractography protocols developed here can only be used via FSL-XTRACT (yet not with other dMRI software), somewhat limiting the overall accessibility of the method.*

We agree that a more quanItative comparison against gold standard tracing data would be ideal. However, there are practical challenges that prohibit such a comparison at this stage: i) Access to data. There are no quantifiable, openly shared, large scale/whole brain tracing data available. The Markov study provided the only openly available weighted connectivity matrices measured by tracers in macaques (Markov, Cereb Cortex 2014), which are only cortico-cortical and do not provide the white matter routes, they only quantify the relative contrast in connection terminals. ii) 2D microscopy vs 3D tractography. The vast majority of tracing data one can find in neuroanatomy labs is on 2D microscopy slices with restricted field of view, which is also the case for the data we had access to for this study. This complicates significantly like-to-like comparisons against 3D whole-brain tractography reconstructions. iii) Quantifiability is even tricky in the case of gold standard axonal tracing, as it depends on nuisance factors, e.g. injection site, injection size, injection uniformity and coverage, which confound the gold-standard measurements, but are not relevant for tractography. For these reasons, a number of high-profile NIH BRAIN CONNECTS Centres (for instance hXps://connects.mgh.harvard.edu/, hXps://mesoscaleconnecIvity.org/) are resourced to address these challenges at scale in the coming years and provide the tools to the community to perform such quantitative comparisons in the future.

In terms of comparison with other approaches, we have performed new tests and detail a response to a similar comment (2) from Reviewer 1.

Finally, our protocols have been FSL-tested, but have nothing that is FSL specific. We cannot speak of performance when used with other tools, but there is nothing that prohibits translation of these standard space protocols to other tools. In fact, the whole idea behind XTRACT was to generate an approach open to external contributions for bundle-specific delineation protocols, both for humans and for non-human species. A number of XTRACT extensions that have been published over the last 5 years for other NHP species (Roumazeilles et al. (2020); Bryant et al. (2020); Wang et al. (2025)) and similar approaches have been used in commercial packages (Boshkovski et al, 2106, ISMRM 2022).

> ***Recommendations To the Authors:***
>
> *(7) Superiority of the FSL-XTRACT approach to delineating cortical-subcortical tracts. The Introduction of the article describes how "Tractography protocols for white matter bundles that reach deeper subcortical regions, for instance the striatum or the amygdala, are more difficult to standardize" due to the size, proximity, complexity, and bottlenecks associated with corticalsubcortical tracts. It would be helpful for the authors to better describe how the analytic approach adopted here overcomes these various challenges. What does the present approach do differently than prior efforts to examine cortical-subcortical connectivity?*

There have not been many prior efforts to standardise cortico-subcortical connecIvity reconstructions, as we overview in the Introduction. As outlined in (Schilling et al. (2020), hXps://doi.org/10.1007/s00429-020-02129-z), tractography reconstructions can be highly accurate if we guide them using constraints that dictate where pathways are supposed to go and where they should not go. This is the philosophy behind XTRACT and all the proposed protocols, which provide neuroanatomical constraints across different bundles. At the same time these constraints are relatively coarse so that they are species-generalisable. We have clarified that in Discussion. The approach we took was to first identify anatomical constraints from neuroanatomy literature for each tract of interest independently, derive and test these protocols in the macaque, and then optimise in an iterative fashion until the protocols generalise well to humans and until, when considering groups of bundles, the generated reconstructions can follow topographical principles known from tract tracing literature. This process took years in order to perform these iterations as meticulously as we could. We have modified the first sections in Methods to reflect this better (3rd paragraph of 1st Methods section), as well as modified the third and second to last paragraphs of the Introduction ("We propose an approach that addresses these challenges…").

> *(8) Relatedly, it is difficult to fully evaluate the utility of the current approach to dissecting cortical-subcortical tracts without a qualitative or quantitative comparison to approaches that already exist in the field. Can the authors show that (or clarify how) the FSL-XTRACT approach is similar to - or superior to - currently available methods for defining cortical-striatal and amygdalofugal tracts (e.g., methods they cite in the Introduction)?"*

From the limited similar approaches that exist, we did perform some comparisons against TractSeg, please see Reply to Comment 2 from Reviewer 1. We have also expanded the relevant text in the introduction to clarify the differences:

"…However, these either uIlise labour-intensive single-subject protocols (22,26), are not designed to be generalisable across species (42, 43), or are based mostly on geometrically-driven parcellaIons that do not necessarily preserve topographical principles of connecIons (40). We propose an approach that addresses these challenges and is automated, standardised, generalisable across two species and includes a larger set of cortico-subcortical bundles than considered before, yielding tractography reconstructions that are driven by neuroanatomical constraints."

> *(9) Future applications of the tractography protocol:*
>
> *It would be helpful for the authors to describe the contexts in which the automated tractography approach developed here can (and cannot) be applied in future studies. Are future applications limited to diffusion data that has been processed with FSL's BEDPOSTX and PROBTRACKX? Can FSL-XTRACT take in diffusion data modelled in other software (e.g., with CSD in mrtrix or with GQI in DSI Studio)? Can the seed/stop/target/exclusion ROIs be applied to whole-brain tractography generated in other software? Integration with other software suites would increase the accessibility of the new tract dissection protocols.*

We have added some text in the Discussion to clarify this point. Our protocols have been FSLtested, but have nothing that is FSL specific. We cannot speak of performance of other tools, but there is nothing that prohibits translaIon of these standard space protocols to other tools. As described before, the protocols are recipes with anatomical constraints including regions the corresponding white matter pathways connect to and regions they do not, constructed with cross-species generalisability in mind. In fact a number of other packages (even commercial) have adopted the XTRACT protocols with success in the past, so we do not see anything in principle that prohibits these new protocols to be similarly adopted.

We cannot comment on the protocols' relevance for segmenIng whole-brain tractograms, as these can induce more false posiIves than tractography reconstructions from smaller seed regions and may require stricter exclusions.

> *(10) It was great to see confirmation that the XTRACT approach can be successfully applied in both high-quality diffusion data from the HCP and in the ON-Harmony data. Given the somewhat degraded performance in the lower quality dataset (e.g., Figure 4A), can the authors speak to the minimum data requirements needed to dissect these new cortical-subcortical tracts? Will the approach work on single-shell, low b data? Is there a minimum voxel resolution needed? Which tracts are expected to perform best and worst in lower-quality data?*

Thank you for these comments, even if we have not really tried in lower (spaIal and angular) resolution data, given the proximity of the tracts considered, as well as the small size of some bundles, we would not recommend lower resolution than those of the UK Biobank protocol. In general, we would consider the UK Biobank protocol (2mm, 2 shells) as the minimum and any modern clinical scanner can achieve this in 6-8 minutes. We hence evaluated performance from high quality HCP to lower quality UK Biobank data, covering a considerable range (scan Ime from 55 minutes down to 6 minutes).

In terms of which tract reconstructions were more reproducible for UKBiobank data, the tracts with lowest correlations across subjects (Figure 4) were the anterior commissure (AC) and the temporal part of the Extreme Capsule (EmC_t), while the highest correlations were for the Muratoff Bundle (MB) and the temporal part of the Striatal Bundle (StB_t). Interestingly, for the HCP data, the temporal part of the Extreme Capsule (EmC_t) and the Muratoff Bundle were also the tracts with the lowest/highest correlations, respectively. Hence, certain tract reconstructions were consistently more variable than others across subjects, which may hint to also being more challenging to reconstruct. We have now clarified these aspects in the corresponding Results section.

> *(11) Anatomical validation of the new cortical-subcortical tracts*
>
> *I really appreciated the use of prior tract tracing findings to anatomically validate the corticalsubcortical tractography outputs for both the cortical-striatal and amygdalofugal tracts. It struck me, however, that the anatomical validation was purely qualitative, focused on the relative positioning or the topographical organization of major connections. The anatomical validation would be strengthened if profiles of connectivity between cortical regions and specific subcortical nuclei or subcortical subdivisions could be quantitatively compared, if at all possible. Can the differential connectivity shown visually for the putamen in Figure 3 be quantified for the tract tracing data and the tractography outputs? Does the amygdalofugal bundle show differential/preferential connectivity across amygdala nuclei in tract tracing data, and is this seen in tractography?*

We appreciate the comment, please see Reply to your comment 6 above. In addiIon to the challenges described there, we do not have access to terminal fields other than in the striatum and these ones are 2D, so we make a qualitaIve comparison of the relevant connecIvity contrasts. We expect that a number of currently ongoing high-profile BRAIN CONNECTS Centres (such as the LINC and the CMC) will be addressing such challenges in the coming years and will provide the tools and data to the community to perform such quanItaIve comparisons at scale.

> *(12) I believe that all visualizations of the macaque and human tractography showed groupaveraged maps. What do these tracts look like at the individual level? Understanding individual-level performance and anatomical variation is important, given the Discussion paragraph on using this method to guide neuromodulation.*

We now demonstrate some representative examples of individual subject reconstructions in Supplementary Figures S6, S7, ranking subjects by the average agreement of individual tract reconstructions to the mean and depicting the 10th percentile, median and 90th percentile of these subjects. We have also shown more results in Author response images 1-2, generated by TractSeg, to indicate how a different bundle segmentation approach would handle individual variability compared to our approach.

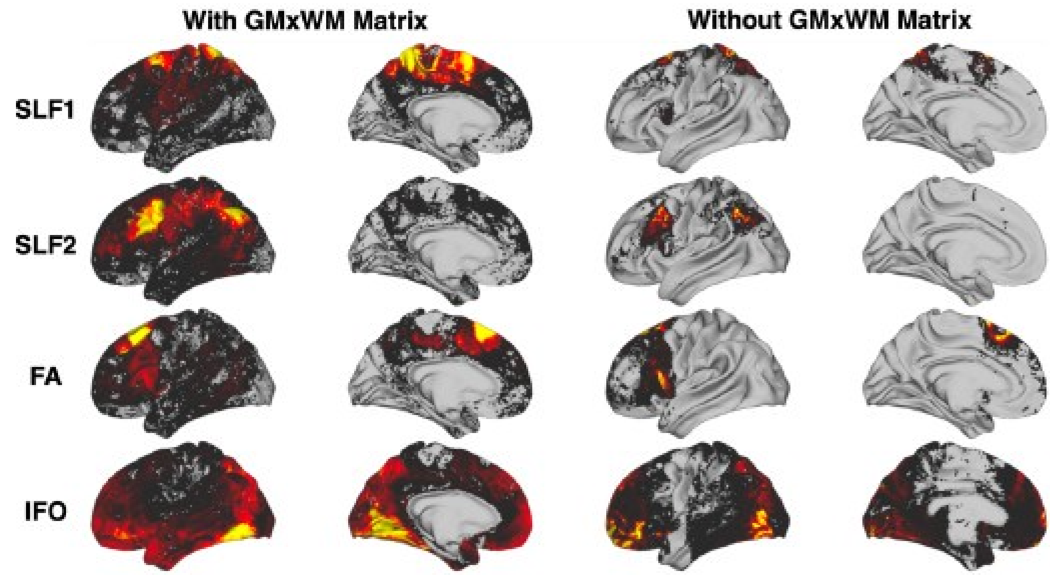
> *(13) Connectivity-based comparisons across species:*
>
> *Figures 5 and 6 of the manuscript show that, as compared to using only cortico-cortical XTRACT tracts, using the full set of XTRACT tracts (with new cortical-subcortical tracts) allows for more specific mapping of homologous subcortical and cortical regions across humans and macaques. Is it possible that this result is driven by the fact that the "connectivity blueprints" for the subcortex did not use an intermediary GM x WM matrix to identify connection patterns, whereas the connectivity blueprints for the cortex did? I was surprised that a whole brain GM x WM connectivity matrix was used in the cortical connectivity mapping procedure, given known problems with false positives etc., when doing whole brain tractography - especially aHer such anatomical detail was considered when deriving the original tracts. Perhaps the intermediary step lowers connectivity specificity and accuracy overall (as per Figure 9), accounting for the poorer performance for cortico-cortical tracts?*

The point is well-taken, however it cannot drive the results in Figures 5 and 6. Before explaining this further, let us clarify the raIonale of using the GMxWM connecIvity matrix, which we have published quite extensively in the past for cortico-cortical connecIons (Mars, eLife 2018 - Warrington, Neuroimage 2020 - Roumazeilles, PLoS Biology 2020 - Warrington, Science Advances 2022 – Bryant, J Neuroscience 2025).

Having established the bodies of the tract using the XTRACT protocols, we use this intermediate step of multiplying with a GM x WM connectivity matrix to estimate the grey matter projections of the tracts. The most obvious approach of tracking towards the grey matter (i.e. simply find where tracts intersect GM) has the problem that one moves through bottlenecks in the cortical gyrus and after which fibres fan out. Most tractography algorithms have problems resolving this fanning. However, we take the opposite approach of tracking from the grey matter surface towards the white matter (GMxWM connectivity matrix), thus following the direction in which the fibres are expected to merge, rather than to fan out. We then multiply the GMxWM tractrogram with that of the body of the tract to identify the grey matter endpoints of the tract. This avoids some of the major problems associated with tracking towards the surface. In fact, using this approach improves connectivity specificity towards the cortex, rather than the opposite. We provide some indicative results here for a few tracts:

Author response image 5.

Connectivity profiles for example cortico-cortical tracts with and without using the intermediary GMxWM matrix. Tracts considered are the Superior Longitudinal Fasciculus 1 (SLF_1), Superior Longitudinal Fasciculus 2 (SLF_2), the Frontal Aslant (FA) and the Inferior Fronto-Occipital Fasciculus (IFO). We see that the surface connectivity patterns without using the GMxWM intermediary matrix are more diffuse (effect of “fanning out” gyral bias), with reduced specificity, compared to whenusing the GMxWM matrix



Tracking to/from subcortical nuclei does not have the same tractography challenges as tracking towards the cortex and in fact we found that using the intermediary GMxWM matrix is less favourable for subcortex (Figure 9), which is why we opted for not using it.

Regardless of how cortical and subcortical connectivity patterns are obtained, the results in Figures 5 and 6 utilise only cortical connectivity patterns. Hence, no matter what tracts are considered (cortico-cortical or cortico-subcortical) to build the connectivity patterns, these results have been obtained by always using the intermediate step of multiplying with the GMxWM connectivity matrix (i.e. it is not the case that cortical features are obtained with the intermediate step and subcortical features without, all of them have the intermediate step applied, as the connectivity patterns comprise of cortical endpoints). Figure 9 is only applicable for subcortical endpoints that play no role in the comparisons shown in Figures 5 and 6. We hope this clarifies this point.

> *(14) Methodological clarifications:*
>
> *The Methods describe how anatomical masks used in tractography were delineated in standard macaque space and then translated to humans using "correspondingly defined landmarks". Can the authors elaborate as to how this translation from macaques to humans was accomplished?*

For a given tract, our process for building a protocol involved looking into the wider anatomical literature, including the standard white matter atlas of Schmahmann and Pandya (2006) and numerous anatomy papers that are referenced in the protocol description, to determine the expected path the tract was meant to take in white matter and which cortical and subcortical regions are connected. This helped us define constraints and subsequently the corresponding masks. The masks were created through the combination of hand-drawn ROIs and standard space atlases. We firstly started with the macaque where tracer literature is more abundant, but, importantly, our protocol definitions have been designed such that the same protocol can be applied to the human and macaque brain. All choices were made with this aspect in mind, hence corresponding landmarks between the two brains were considered in the mask definition (for instance “the putamen”, “a sub-commissural white matter mask”, the “whole frontal pole” etc, as described in the protocol descriptions).

The protocols have not been created by a single expert but have been collated from multiple experts (co-authors SA, SW, DF, KB, SH, SS drove this aspect) and the final definitions have been agreed upon by the authors.

> *(15) The article heavily utilizes spatial path distribution maps/normalized path distributions, yet does not describe precisely what these are and how they were generated. Can the authors provide more detail, along with the rationale for using these with Pearson's correlations to compare tracts across subjects (as opposed to, e.g., overlap sensitivity/specificity or the Jaccard coefficient)?*

We have now clarified in text how these plots are generated, particularly when compared using correlation values. We tried Jaccard indices on binarized masks of the tracts and these gave similar trends to the correlations reported in Figure 4 (i.e. higher similarities within that across cohorts). We however feel that correlations are better than Jaccard indices, as the latter assume binary masks, so they focus on spatial overlap ignoring the actual values of the path distributions, we hence kept correlations in the paper.

> ***Reviewing Editor Comments***
>
> *"The reviewers had broadly convergent comments and were enthusiastic about the work. As further detailed by Reviewer 3 (see below), if the authors choose to pursue revisions, there are several elements that have the potential to enhance impact."*

Thank you, we have replied accordingly and aimed to address most of the comments of the Reviewers.

> *"Comparison to existing methods. How does this approach compare to other approaches cited by the authors?"*

Please see replies to Comment 2 of Reviewer 1 and Comment 7 of Reviewer 2. Briefly, we have now generated new results and clarified aspects in the text.

> *"Minimum data requirements. How broadly can this approach be used across scan variation? How does this impact data from individual participants? Displaying individual participants may help, in addition to group maps."*

Please see replies to Comment 10 of Reviewer2 on minimum data requirements and individual parIcipants, as well as to Comment 3 of Reviewer 1 on the actual groups considered. Briefly, we have generated new figures and regenerated results using UKBiobank data.

> *Softare. What are the sofware requirements? Is the approach interoperable with other methods?"*

Please see Reply to Comment 9 of Reviewer 2. Our protocols can be used to guide tractography using other types of data as they comprise of guiding ROIs for a given tract. So, although we have not tested them beyond FSL-XTRACT, we believe they can be useful with other tractography packages as well, as there is nothing FSL-specific in these anatomically-informed recipes.

> *"Comparisons with tract tracing. To the degree possible, quantitative comparisons with tract tracing data would bolster confidence in the method."*

Please see Replies to Comments 6 and 11 of Reviewer 2. Briefly, we appreciate the comment and it is something we would love to do, but there are no data readily available that would allow such quanItaIve comparison in a meaningful way. This is a known challenge in the tractography field, which is why NIH has invested in two 5 year Centres to address it. Our approach will provide a solid starIng point for opImising and comparing further cortico-subcortical tractography reconstructions against microscopy and tracers in the same animal and at scale.

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